

Project 3 - March Cohort

Bharat Paliwal

Project 3 - Travel RAG Chatbot Project

AI-Powered Travel Assistant with Governance & LLMOps

Ground Rules : Important

Ground rules for setting up an Azure Resources for this project

- Utilize the default resource group for all your tasks

NOTE: To optimize resource management, use the existing resource group i.e default_resource_group for all tasks in the project

- Avoid creating multiple resources and resource groups to ensure consistency, simplify monitoring, and reduce overhead
- Select Region = **East US** for the Azure AI Foundry
- Pricing tier: **Standard S0** pricing tier
- If **Azure AI Foundry** has already been deployed, the setup process for it can be skipped, and the focus can shift directly to deploying the model(s)
- **Allow 10 to 15 minutes** for resources to become fully operational following its creation

Instructions

- All screenshots should be clear, high-quality, and legible, ensuring that details are easily visible
- Wherever applicable, the learner's account details should be visible to verify ownership and authenticity
- **Red boxes** should be correctly placed to highlight the key steps in each screenshot, ensuring clarity and guiding the viewer through the process
- The overall structure and logical flow of the process must be maintained by placing each screenshot under the appropriate slide headers
- Learners should avoid skipping steps or presenting them in an incorrect order, as this can disrupt the coherence of the submission

Step 1: Define Business context and objective

1.1 Problem Statement

- WanderNest Travels faces frequent customer complaints, booking abandonment, and platform trust issues due to unresolved travel-related queries.
- Travel information changes rapidly due to flight disruptions, visa requirements, baggage rules, airline changes, hotel availability, and geopolitical events.
- Static FAQs and generic chatbots cannot reliably address these real-time, context-specific queries.
- Human agents currently spend **10–15 minutes per query**, creating customer frustration, operational bottlenecks, and revenue loss.
- Policy-related confusion, particularly around **baggage allowances and cancellations**, contributes to customer drop-offs.

1.2 Business Context

- WanderNest Travels is a **mid-sized online travel agency serving 500K+ users annually**.
- The company processes **62,000+ annual bookings** from key Indian hubs: BLR, DEL, BOM
- Major international destinations include: LON, JFK, DXB, BKK
- Customer inquiries span fragmented information sources, including **live APIs, news feeds, internal policies, and user itineraries**.
- The business objective is to reduce customer confusion and operational cost by shifting **first-line responses** to an RAG-powered chatbot.
- The solution aims to improve query deflection, customer satisfaction, operational agility, and retention within the WanderNest ecosystem.

1.3 Data Description

The RAG chatbot uses two categories of information:

1. Static Knowledge Base

- Airline policies
- Ticket refund policies
- Travel policies
- Routes
- FAQs
- Policy and fare information

2. Dynamic / Updated Information

- Weather-related disruptions
- Personnel strikes
- Airport policy changes
- Immigration policy changes
- Latest news and other relevant travel information
- Time-sensitive events such as fare increases and visa-policy shifts

Purpose: The data enables the chatbot to combine established travel policies with current information to provide context-aware responses to customer queries.

1.4 Data Source

The project draws information from multiple sources described in the business requirements:

- **Internal/static documents:** Policies, fares, routes, FAQs, and ticket refund policies.
- **Live APIs:** Used as one of the fragmented sources available to agents for current information.
- **News feeds:** Provide information on evolving travel events and disruptions.
- **Post-deployment data feeds:** Provide dynamic updates that can be integrated into the RAG system.
- **Government advisories:** Used for source-verified health, security, and regulatory information.
- **User itineraries:** Provide context relevant to individual customer queries.

1.5 Solution methodology

Solution Methodology — Azure-Based Dynamic RAG

1. Data Ingestion & Storage

- Store static documents such as **airline policies, refund policies, routes, FAQs and fares** in **Azure Blob Storage**.
- Ingest dynamic updates such as weather events, strikes, airport/immigration policy changes and relevant news through available data feeds/APIs.
- Maintain separate static and dynamic knowledge sources to support timely updates.

2. Document Processing

- Use **Azure AI Services** to process and extract relevant information from documents where required.
- Chunk documents into searchable content and retain relevant metadata such as policy/topic/source.

1.5 Solution methodology (contd.)

3. Knowledge Indexing & Retrieval

- Use **Azure AI Search** to create the RAG knowledge index.
- Support keyword/vector or hybrid retrieval to identify the most relevant policy and travel information for each user query.
- Azure AI Search is specifically designed to support RAG through text, vector and hybrid search.

4. Query Understanding & Context Retrieval

- User submits a question such as:
“Is the BLR-LON route impacted by the new EU baggage rules?”
- The application searches Azure AI Search for relevant policy, route and current-event information.
- Retrieved information is passed as grounding context to the LLM.

1.5 Solution methodology (contd.)

5. Response Generation

- Use **Azure OpenAI / Microsoft Foundry with GPT-5.4-mini** as the generation model.
- GPT-5.4-mini generates a response based on the retrieved context rather than relying solely on its pretrained knowledge.
- Responses should include the relevant source/policy references where available to improve transparency and reduce unsupported answers.

6. Dynamic Knowledge Updates

- Continuously update the knowledge base when new baggage, visa, disruption, fare or other travel information becomes available.
- Re-index updated content in **Azure AI Search**, allowing the chatbot to use the latest available information without retraining the LLM.

1.5 Solution methodology (contd.)

7. Application & Deployment

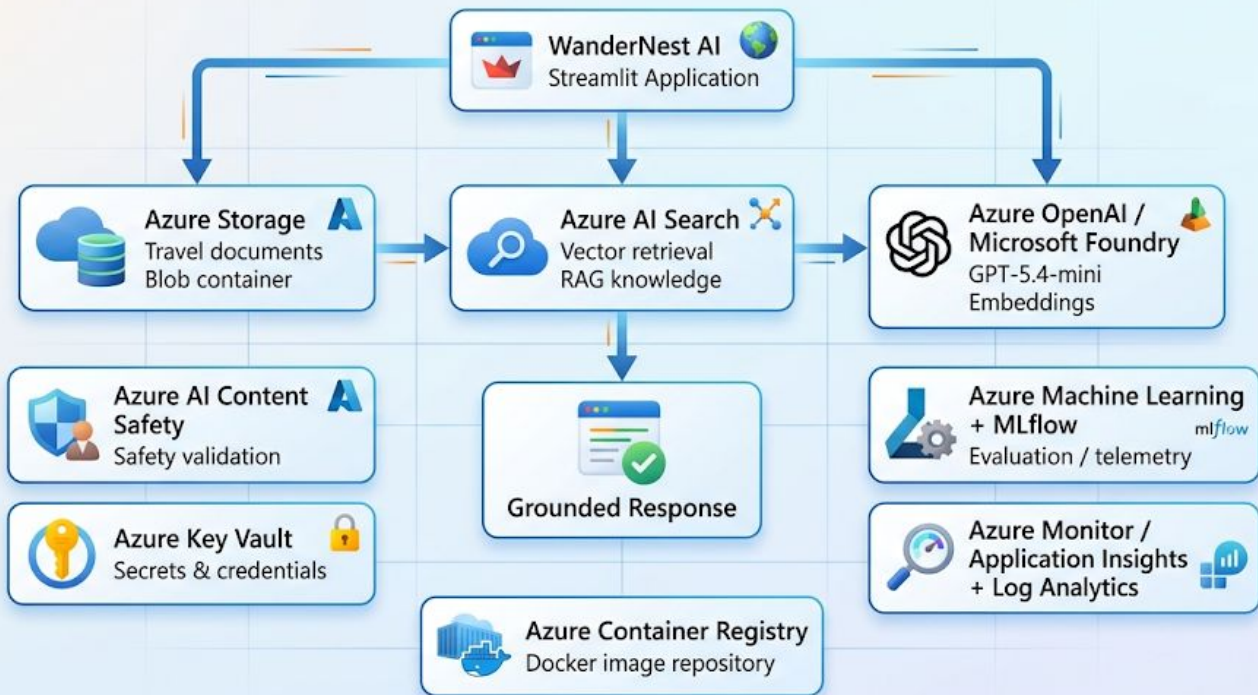
- Expose the RAG chatbot through an application/API hosted on **Azure App Service** or an equivalent Azure application service.
- Use Azure-managed authentication and service-to-service access where applicable. Microsoft provides an Azure architecture combining Azure OpenAI, Azure AI Search and App Service for RAG applications.

8. LLMOps & Monitoring

- Monitor chatbot responses, retrieval quality, latency and failures.
- Evaluate responses against the available knowledge base to identify incorrect or unsupported answers.
- Maintain versioned knowledge sources and update the index as policies and travel information change.

Step 2: Provision Azure Resources

WanderNest AI Architecture



Azure Foundry Service

2.1 Azure AI Foundry Resource creation

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Home > Microsoft Foundry

 Microsoft Foundry | Foundry ...

×

 Search

×

«

-  Overview
-  All resources
- Use with Foundry
 -  **Foundry**
 -  AI Hubs
 -  Azure OpenAI
 -  AI Search
- > More services
- > Classic AI services
- > Help

 Create

 Manage deleted resources

 Manage view ▾

 Refresh

 Export to CSV

 Open query

|

 Assign tags

+

Add to service group

 Group by none ▾

Create new Microsoft Foundry

Subscription equals all

Type equals all

Resource Group equals all ×

Location equals all ×

+ Add filter

☐	Name ↑		Kind	Location	Custom Domai...	Pricing tier	Status	Created date
☐	 bsp-project2-resource	...	AI Services	East US 2	bsp-project2-res...	S0	Succeeded	2026-07-05T03:...



Create a Foundry resource



[Basics](#) [Storage](#) [Inbound Networking](#) [Outbound Networking](#) [Identity](#) [Encryption](#) [Tags](#) [Review + create](#)

Design, customize, and manage AI apps and agents at scale. [Learn More](#)

Instance Details

Select the subscription to manage deployed resources and costs. Use resources groups like folders to organize and manage all your resources.

Subscription *	npglb2cms-1773207764316
Resource group *	marchcohortbpaliwal
	Create new
Name *	bsp-project3-resource
Region	(US) East US

[View full pricing details](#)

Your first project

Foundry organizes development artifacts into projects— containers for managing permissions, monitoring, costs, and more. When you create a resource, a project is also created.This "default" project has more capabilities than projects that you create later. [Learn more](#).

Default project name *	proj-wandernest
------------------------	--

Content Review Policy

[Previous](#)[Next](#)[Review + create](#)[Give feedback](#)

Create a Foundry resource



- Basics
- Storage
- Inbound Networking
- Outbound Networking
- Identity
- Encryption
- Tags
- Review + create

 [View automation template](#)

Basics

Subscription	npglb2cms-1773207764316
Resource group	marchcohortbpaliwal
Name	bsp-project3-resource
Region	East US
Default project name	proj-wanderlust

Storage

Key Vault	
Application Insights	
Storage for Agents service	0 item(s)
Storage Account (preview)	

Inbound Networking

Inbound Access	All networks, including the internet, can access this resource.
----------------	---

Outbound Networking

Home



AI Foundry Create-20260803172643 | Overview

Deployment



Search



Delete



Cancel



Redeploy



Download



Refresh



Overview



Inputs



Outputs



Template

Deployment is in progress



Deployment name : AI Foundry Create-20260803172643

Subscription : [npglb2cms-1773207764316](#)

Resource group : [marchcohortbpaliwal](#)

Start time : 8/3/2026, 5:27:33 PM

Correlation ID : c6e4f658-d0dd-41ba-99bf-dbb2c435c379

Deployment details

Resource	Type	Status	Operation details
bsp-project3-resource	Microsoft.CognitiveServices/acc	Created	Operation details



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Home



AIFoundryCreate-20260803172643 | Overview

Deployment



Search



Delete



Cancel



Redeploy



Download



Refresh



Overview



Inputs



Outputs



Template



Your deployment is complete



Deployment name : AIFoundryCreate-20260803172643

Subscription : [npglb2cms-1773207764316](#)

Resource group : [marchcohortbpaliwal](#)

Start time : 8/3/2026, 5:27:39 PM

Correlation ID : c6e4f658-d0dd-41ba-99bf-dbb2c435c379



Deployment details



Next steps

[Go to resource](#)



Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)



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2.2 Azure AI Foundry Interface

(Put the screenshot of the Azure AI Foundry user interface here)

Azure services


Create a resource


Foundry


AI Search (Foundry IQ)


Resource groups


Quickstart Center


Kubernetes services


Virtual machines


App Services


Storage accounts


More services

Resources

Recent Favorite

Name	Type	Last Viewed
 bsp-project3-resource	Foundry	a minute ago
 bsp-project2-resource	Foundry	3 weeks ago

See all

Navigate

 Subscriptions

 Resource groups

 All resources

 Dashboard

Tools


Microsoft Learn 
Learn Azure with free online training from Microsoft


Azure Monitor
Monitor your apps and infrastructure


Microsoft Defender for Cloud
Secure your apps and infrastructure


Cost Management
Analyze and optimize your cloud spend for free

Home



bsp-project3-resource

☆ ☆ ...



Show platform metrics for this AI resource

Best practices for managing AI resources



 Search



 Go to Foundry portal

 Delete

 Overview

-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer
- > Resource Management
- > Security
- > Monitoring
- > Automation
- > Help

^ Essentials

[JSON View](#)

Resource group (move) :	marchcohortbpaliwal	API Kind : AIServices
Subscription (move) :	npglb2cms-1773207764316	Location : eastus
Subscription ID :	a0e7e508-3870-4e84-b3d3-d1a183ee5089	Status : Succeeded
Endpoints :	Click here to view endpoints	
Tags (edit) :	Add tags	



Build apps and agents in Foundry

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Go to Foundry portal



Welcome, Bharat



Use a model

Manage your existing endpoints or deploy a new model.

[View deployments](#)

Build an agent

Create an AI agent by configuring instructions, tools, and knowledge without writing code.

[Start building](#)

Code an agent

Define and deploy an agent in code using Microsoft Agent Framework for full control and customization.

[Start coding](#)

API key

.....

Project endpoint ⓘ

<https://bsp-project3-resource.services.ai.azure.com/api/>

Azure OpenAI endpoint

<https://bsp-project3-resource.openai.azure.com/openai/>

Model selection



Test in playground

Run prompts and see responses.



Explore models

Browse available models.



Compare models

Review benchmarks and capabilities.



claude-opus-5
Anthropic



gpt-5.6-sol
OpenAI



MAI-Image-2.5
Microsoft



DeepSeek-V4-Pro
DeepSeek



Foundry Model Deployment

In this section, provide details of the models used in the project, its function and the reason for selecting the model

Model Selection

Model	Function	Reason for Selection
GPT-5.4-mini	Response Generation — generates the final answer using the user's question and retrieved travel-document context	Strong reasoning and response quality with good latency and cost efficiency for a RAG chatbot
text-embedding-3-small	Semantic Retrieval — converts document chunks and user queries into embeddings for vector search in Azure AI Search	Efficient and cost-effective embeddings suitable for semantic search over the travel knowledge base
GPT-5.4-mini + text-embedding-3-small (RAGAS)	RAG Evaluation — evaluates generated answers using RAGAS metrics for response quality, relevance, faithfulness, and retrieval quality	Provides an automated, repeatable evaluation framework using the same Azure OpenAI model family as the application



Agents



Deployments



Fine-tune



Services



Tools



Knowledge



Memory



Data



Evaluations



Guardrails

Deployments

Deployed models

Models

[Preview](#)

Batch jobs

Serverless deployments

Managed compute deployments

Search



PTU Calculator

Deploy

Name	Model	Version	Deployment status	Deployment type	Created on ↓	Created by
text-embedding-3-small	text-embed...	1	Succeeded	Global Stan...	8/7/26, 9:01:...	Bharat_1776...
gpt-5.4-mini	gpt-5.4-mini	2026-03-17	Succeeded	Global Stan...	8/7/26, 7:31:...	Bharat_1776...

1-2 of 2

[< Prev](#) [Next >](#)

gpt-5.4-mini



v:2026-03-17 Succeeded

Deployed 8/7/2026, 7:31:38 PM

[Open in playground](#)[Edit](#)[Delete](#)

Send a request to this model.

Project endpoint

<https://bsp-project3-resource.servi>

API Key

.....

Call this model

Language Python

...



```
1 from openai import OpenAI
2 from azure.identity import DefaultAzureCr
3
```



Skip setup with VS Code for the Web

Start coding with tools, credentials, and dependencies already set up.



Model Deployment flows

Home

 **bsp-project3-resource**
Foundry

 ... 

Show platform metrics for this AI resource

Best practices for managing AI resources



 Search





 Go to Foundry portal

 Delete

-  Overview
-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer
- > Resource Management
- > Security
- > Monitoring
- > Automation
- > Help

^ Essentials JSON View

Resource group [\(move\)](#) : [marchcohortbpaliwal](#)

Subscription [\(move\)](#) : [npglb2cms-1773207764316](#)

Subscription ID : a0e7e508-3870-4e84-b3d3-d1a183ee5089

Endpoints : [Click here to view endpoints](#)

Tags [\(edit\)](#) : [Add tags](#)

API Kind : AIServices

Location : eastus

Status : Succeeded



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https://bsp-project3-resource.services.ai.azure.com/api/

Azure OpenAI endpoint

https://bsp-project3-resource.openai.azure.com/openai/

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claude-opus-5
Anthropic



gpt-5.6-sol
OpenAI



MAI-Image-2.5
Microsoft



DeepSeek-V4-Pro
DeepSeek





Agents

Deployments

Fine-tune

Services

Tools

Knowledge

Memory

Data

Evaluations

Guardrails

Deployments

[Deployed models](#)[Models](#)[Preview](#)[Batch jobs](#)[Serverless deployments](#)[Managed compute deployments](#)

Search

PTU Calculator

Deploy



Deploy your first model

After you deploy a model, you can try it out in the playground.



- Overview
- Models
- Agents
- Tools
- Services
- Solution templates

Models (196)

Compare models View leaderboard

Search Sort by Featured

- Availability
 - All models
 - Available in my project
- Collections
- Source
- Supported features
- Inference tasks
- Deployment options
- Deployment SKU
- Region
- Fine-tuning methods
- Lifecycle
- Domain

 gpt-5.6-sol Chat completion, Responses	 gpt-5.6-luna Chat completion, Responses	 gpt-5.6-terra Chat completion, Responses
 claude-opus-5 Messages	 claude-sonnet-5 Messages	 Cohere-command-a-plus-05-20... Chat completion
 MAI-Voice-2 Text to speech, Audio generation Microsoft	 MAI-Transcribe-1.5 Automatic speech recognition, Speech to Microsoft	 MAI-Image-2.5 Text to image Microsoft
 MAI-Image-2.5-Flash Text to image Microsoft	 claude-opus-4-8 Messages	 grok-4.3 Chat completion, Responses
 DeepSeek-V4-Pro Chat completion	 DeepSeek-V4-Flash Chat completion	 Kimi-K2.6 Chat completion
 MAI-Image-2e Text to image Microsoft	 grok-4-20-reasoning Chat completion, Responses	 zai-org--glm-5.2-fp8 Chat completion Hugging Face
 grok-4-20-non-reasoning Chat completion, Responses	 qwen--qwen3.6-27b Chat completion Hugging Face	 MAI-Voice-1 Text to speech, Audio generation Microsoft



- Overview
- Models**
- Agents
- Tools
- Services
- Solution templates

Sort by Featured

- Availability
- All models

Available in my project ⓘ
- Collections
- Source
- Supported features
- Inference tasks
- Deployment options
- Deployment SKU
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- Fine-tuning methods
- Lifecycle
- Domain
- Industry

Search results for "mini" (14)



o4-mini
Chat completion, Responses



o3-mini
Chat completion, Responses



gpt-5-mini
Chat completion, Responses



gpt-5.1-codex-mini
Responses



gpt-4.1-mini
Chat completion, Responses



Phi-4-mini-instruct
Chat completion



gpt-4o-mini
Chat completion, Responses

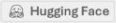


gpt-5.4-mini
Chat completion, Responses



Phi-4-mini-reasoning
Chat completion



cohere--north-mini-code-1.0
Chat completion


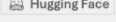


cohere--north-mini-code-1.0
Chat completion




Ministral-3B
Chat completion



sentence-transformers--all-min...
Embeddings




mistralai--ministral-3-3b-instru...
Chat completion




- Overview
- Models
- Agents
- Tools
- Services
- Solution templates

gpt-5.4-mini

Deploy

- Details
- Deployments
- Benchmarks
- Responsible AI
- License

GPT-5.4-mini is a compact, cost-efficient model designed for reliable performance across high-volume, everyday AI workloads.

Azure OpenAI

Direct from Azure

Version: 2026-03-17

Direct from Azure models

Direct from Azure models are a select portfolio curated for their market-differentiated capabilities:

- Secure and managed by Microsoft: Purchase and manage models directly through Azure with a single license, consistent support, and no third-party dependencies, backed by Azure's enterprise-grade infrastructure.
- Streamlined operations: Benefit from unified billing, governance, and seamless PTU portability across models hosted on Azure - all part of Microsoft Foundry.
- Future-ready flexibility: Access the latest models as they become available, and easily test, deploy, or switch between them within Microsoft Foundry; reducing integration effort.
- Cost control and optimization: Scale on demand with pay-as-you-go flexibility or reserve PTUs for predictable performance and savings.

Learn more about [Direct from Azure models](#).

Key capabilities

About this model

GPT-5.4-mini is a compact, cost-efficient model designed for reliable performance across high-volume, everyday AI workloads.

Key model capabilities

- Most capable OpenAI frontier model for professional work
- Stronger reasoning for complex, multi-step tasks

Quick facts

Model provider	Azure OpenAI
Type	Chat completion, Responses
Lifecycle	Generally Available
Input type	text, image
Output type	text
Context window	400k
Token limits	128k output
Pricing	View pricing
Availability	View availability

Model ID

azureml://registries/azure-openai/models/gpt-5.4-mini/versions/2026-03-17

gpt-5.4-mini

Details Deployments Benchmarks Responsible AI License

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 Direct from Azure

Version: 2026-03-17

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- Stronger reasoning for complex, multi-step tasks

Deploy

Default settings

Set to global standard and default quota.

Custom settings

Set your own SKU, quota, PTU, spillover, and guardrails.

Responses	
Lifecycle	Generally Available
Input type	text,image
Output type	text
Context window	400k
Token limits	128k output
Pricing	View pricing
Availability	View availability

Model ID

azureml://registries/azure-openai/
models/gpt-5.4-mini/versions/2026-03-17



- Agents
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- Data
- Evaluations
- Guardrails

← gpt-5.4-mini

Save as agent

Compare models

Playground

Details

Monitor

Evaluation

Model: gpt-5.4-mini

Global Standard deployment

Instructions

^

You are an AI assistant that helps people find information.

Tools

^

Connect tools to your models for faster access to key information and improved efficiency. [Learn more](#) about best practices. Some tools will require creating an agent.

Web search uses Grounding with Bing, which has additional costs and terms: [terms of use](#) and [privacy statement](#). Customer data will flow outside the Azure compliance boundary. [Learn more here](#).

Web search

Add

Upload files

Knowledge

^

Chat

Call model

Send a request to this model.

Project endpoint

https://bsp-project3-resource.services.ai

API Key

.....

Call this model

Language Python Authentication method Entra ID authentication

```
1 from openai import OpenAI
2 from azure.identity import DefaultAzureCredential, get_bearer_token_provider
3
4 endpoint = "https://bsp-project3-resource.services.ai.azure.com/openai/v1"
5 deployment_name = "gpt-5.4-mini"
6 token_provider = get_bearer_token_provider(DefaultAzureCredential(), "https://ai.azure.com")
7
8 client = OpenAI(
9     base_url=endpoint,
10    api_key=token_provider
11 )
12
13 response = client.responses.create(
14     model=deployment_name,
15     input="What is the capital of France?"
```

Skip setup with VS Code for the Web

Start coding with tools, credentials, and dependencies already set up.

- Agents
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- Services
- Tools
- Knowledge
- Memory
- Data
- Evaluations
- Guardrails

Deployments

Deployed models Models Preview Batch jobs

Serverless deployments Managed compute deployments

Search

PTU Calculator

Deploy

Name	Model	Version	Deployment status	Deployment type	Created on ↓	Created by
gpt-5.4-mini	gpt-5.4-mini	2026-03-17	Succeeded	Global Stan...	8/7/26, 7:31:...	Bharat_1776...

1-1 of 1

Prev Next

gpt-5.4-mini

v:2026-03-17 Succeeded
Deployed 8/7/2026, 7:31:38 PM

Open in playground Edit Delete

Send a request to this model.

Project endpoint

https://bsp-project3-resource.servi

API Key

.....

Call this model

Language Python ...

```
1 from openai import OpenAI
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```



Skip setup with VS Code for the Web

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Agents

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Deployed models[Models](#)[Preview](#)[Batch jobs](#)**Serverless deployments**

Managed compute deployments

Search

PTU Calculator

Name	Model	Version	Deployment status	Deployment type	Created on ↓
gpt-5.4-mini	gpt-5.4-mini	2026-03-17	Succeeded	Global Stan...	8/7/26, 7:31:...

1-1 of 1

[< Prev](#) [Next >](#)**Deploy** ▾

Deploy a base model

Deploy a fine-tuned model

gpt-5.4-miniv:2026-03-17 **Succeeded**

Deployed 8/7/2026, 7:31:38 PM

[Open in playground](#)[Edit](#)[Delete](#)

Send a request to this model.

Project endpoint

<https://bsp-project3-resource.servi>

API Key

.....

Call this model

Language Python ▾ ...

```
1 from openai import OpenAI
2 from azure.identity import DefaultAzureCr
3
```

**Skip setup with VS Code for the Web**

Start coding with tools, credentials, and dependencies already set up.

Overview

Models

Agents

Tools

Services

Solution templates

Models (19)

embedding



Sort by Featured



Compare models

View leaderboard

Availability

All models

Available in my project ⓘ

Collections

Source

Supported features

Inference tasks

Deployment options

Deployment SKU

Region

Fine-tuning methods

Lifecycle

Domain

Search results for "embedding" (19)



qwen--qwen3-embedding-0.6b

Embeddings

Hugging Face



text-embedding-ada-002

Embeddings



qwen--qwen3-embedding-8b

Embeddings

Hugging Face



text-embedding-3-large

Embeddings



text-embedding-3-small

Embeddings



sentence-transformers--embed...

Embeddings

Hugging Face



embed-v-4-0

Embeddings, Summarization



Cohere-embed-v3-multilingual

Embeddings



nvidia--nemotron-3-embed-1b-...

Embeddings

Hugging Face



nvidia--nemotron-3-embed-8b-...

Embeddings

Hugging Face



nvidia--nemotron-3-embed-1b-...

Embeddings

Hugging Face



sentence-transformers--all-min...

Embeddings

Hugging Face



maiya--multilingual-e5-small

Embeddings

Hugging Face



intfloat--e5-base

Embeddings

Hugging Face



intfloat--e5-large-v2

Embeddings

Hugging Face



snowflake--snowflake-arctic-e...

Embeddings

Hugging Face



sentence-transformers--msmar...

Embeddings

Hugging Face



thenlper--gte-large

Embeddings

Hugging Face





text-embedding-3-small

Deploy

[Details](#) [Deployments](#) [Benchmarks](#) [Responsible AI](#) [License](#)

Text-embedding-3 series models are the latest and most capable embedding model from OpenAI.



Azure OpenAI



Direct from Azure

Version: 1

Direct from Azure models

Direct from Azure models are a select portfolio curated for their market-differentiated capabilities:

- Secure and managed by Microsoft: Purchase and manage models directly through Azure with a single license, consistent support, and no third-party dependencies, backed by Azure's enterprise-grade infrastructure.
- Streamlined operations: Benefit from unified billing, governance, and seamless PTU portability across models hosted on Azure - all part of Microsoft Foundry.
- Future-ready flexibility: Access the latest models as they become available, and easily test, deploy, or switch between them within Microsoft Foundry; reducing integration effort.
- Cost control and optimization: Scale on demand with pay-as-you-go flexibility or reserve PTUs for predictable performance and savings.

Learn more about [Direct from Azure models](#).

Key capabilities

About this model

The provider has not supplied this information.

Key model capabilities

The provider has not supplied this information.



Quick facts

Model provider	Azure OpenAI
Type	Embeddings
Lifecycle	Generally Available

Input type	text
Output type	embeddings

Context window	8191
----------------	------

Pricing [View pricing](#)

Availability [View availability](#)

Model ID

```
azureml://registries/azure-openai/  
models/text-embedding-3-small/v  
ersions/1
```

Others who deployed this model also used



text-embedding-3-small

Deploy

Details

Deployments

Benchmarks

Responsible AI

License

Text-embedding-3 series models are the latest and most capable embedding model from OpenAI.



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Learn more about [Direct from Azure models](#).

Key capabilities

About this model

The provider has not supplied this information.

Key model capabilities

The provider has not supplied this information.



Default settings

Set to global standard and default quota.

Custom settings

Set your own SKU, quota, PTU, spillover, and guardrails.

Lifecycle

Generally Available

Input type

text

Output type

embeddings

Context window

8191

Pricing

[View pricing](#)

Availability

[View availability](#)

Model ID

```
azureml://registries/azure-openai/
models/text-embedding-3-small/v
ersions/1
```



Others who deployed this model also used

-  Agents
-  **Deployments**
-  Fine-tune
-  Services
-  Tools
-  Knowledge
-  Memory
-  Data
-  Evaluations
-  Guardrails

← **text-embedding-3-small**

Playground

Details

Monitor

Evaluation

Edit

Delete

Request quota 

Endpoint

https://bsp-project3-resource.services.ai.azure.com

Key

.....

Deployment info

Name	Priority processing
text-embedding-3-small	N/A
Deployment type	Created on
GlobalStandard	Aug 7, 2026 at 9:01 PM
Provisioning state	Last modified on
 Succeeded	Aug 7, 2026 at 9:01 PM
Version upgrade policy	Create By
OnceNewDefaultVersionAvailable	Bharat_1776808805584@npglb2cms.onmicrosoft.com
Spillover deployment	Modified by
N/A	Bharat_1776808805584@npglb2cms.onmicrosoft.com

Tokens per Minute Rate Limit	Requests per Minute Rate Limit
1250000	7500

Model Deployment

(Put the screenshot of the model selected in the “models + deployments” section here)



Agents

Deployments

Fine-tune

Services

Tools

Knowledge

Memory

Data

Evaluations

Guardrails

← text-embedding-3-small

Playground

Details

Monitor

Evaluation

Edit

Delete

Request quota



Endpoint

https://bsp-project3-resource.services.ai.azure.com



Key

.....



Deployment info

Name

text-embedding-3-small

Priority processing

N/A

Deployment type

GlobalStandard

Created on

Aug 7, 2026 at 9:01 PM

Provisioning state

🟢 Succeeded

Last modified on

Aug 7, 2026 at 9:01 PM

Version upgrade policy

OnceNewDefaultVersionAvailable

Create By

Bharat_1776808805584@npglb2cms.onmicrosoft.com

Spillover deployment

N/A

Modified by

Bharat_1776808805584@npglb2cms.onmicrosoft.com

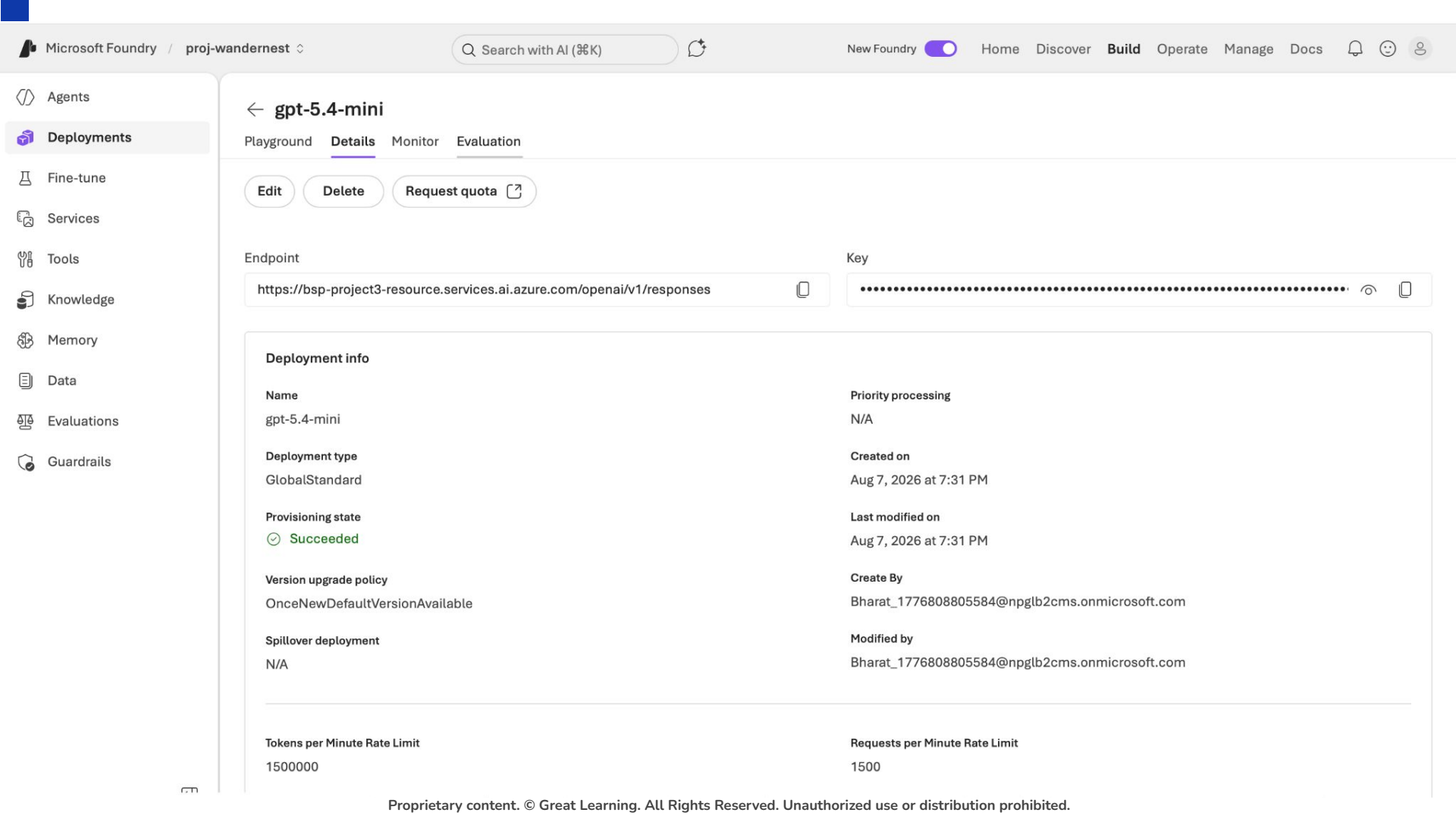
Tokens per Minute Rate Limit

1250000

Requests per Minute Rate Limit

7500





Agents

Deployments

Fine-tune

Services

Tools

Knowledge

Memory

Data

Evaluations

Guardrails

← gpt-5.4-mini

Playground **Details** Monitor Evaluation

Edit

Delete

Request quota

Endpoint

https://bsp-project3-resource.services.ai.azure.com/openai/v1/responses

Key

.....

Deployment info

Name

gpt-5.4-mini

Deployment type

GlobalStandard

Provisioning state

✓ Succeeded

Version upgrade policy

OnceNewDefaultVersionAvailable

Spillover deployment

N/A

Priority processing

N/A

Created on

Aug 7, 2026 at 7:31 PM

Last modified on

Aug 7, 2026 at 7:31 PM

Create By

Bharat_1776808805584@npglb2cms.onmicrosoft.com

Modified by

Bharat_1776808805584@npglb2cms.onmicrosoft.com

Tokens per Minute Rate Limit

1500000

Requests per Minute Rate Limit

1500

All services > Microsoft Foundry

Microsoft Foundry | Content safety

Search

Overview

All resources

> Use with Foundry

> More services

> Classic AI services

Help

Support + Troubleshooting

[+ Create](#) [Manage deleted resources](#) [Manage view](#) [Refresh](#) [Export to CSV](#) [Open query](#) [Assign tags](#) [Delete](#) [Group by none](#)

Filter for any field...

Subscription equals all

Type equals all

Resource Group equals all

Location equals all

+ Add filter



No Content safety to display

An AI service that detects unwanted contents.

[+ Create](#)[Learn more](#)

Showing 1 - 0 of 0. Display count: auto

[Give feedback](#)

Create Content Safety

- Basics
- Network
- Identity
- Tags
- Review + create

Machine learning assisted moderation APIs that detect material that is potentially offensive, risky, or otherwise undesirable, to assure the contents in community is safe.

[Learn more](#)

Project Details

Subscription * ⓘ

npglb2cms-1773207764316

Resource group * ⓘ

marchcohortbpaliwal

[Create new](#)

Instance Details

Region ⓘ

(US) East US

Name * ⓘ

bsprrj3safety

Pricing tier * ⓘ

Free F0

[View full pricing details](#)



Create Content Safety



- Basics
- Network
- Identity
- Tags
- Review + create



[View automation template](#)

Basics

Subscription	npglb2cms-1773207764316
Resource group	marchcohortbpaliwal
Region	East US
Name	bsprrj3safety
Pricing tier	Free F0

Network

Type	All networks, including the internet, can access this resource.
------	---

Identity

Identity type	None
---------------	------

Previous

Next

Create



Give feedback

All services

ContentSafetyCreate-20260816180603 | Overview

Deployment

Search

×

«

🗑️ Delete

🚫 Cancel

🔄 Redeploy

⬇️ Download

🔄 Refresh

- Overview
- Inputs
- Outputs
- Template

Deployment is in progress

Deployment name : ContentSafetyCreate-20260816180603

Subscription : [npglb2cms-1773207764316](#)

Resource group : [marchcohortbpaliwal](#)

Start time : 8/16/2026, 6:06:59 PM

Correlation ID : f07cceb8-651b-4e47-8dbe-ea9840c05d75

Deployment details

Resource	Type	Status	Operation details
There are no resources to display.			



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All services

ContentSafetyCreate-20260816180603 | Overview

Deployment

Search

×

«

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete



Deployment name : ContentSafetyCreate-20260816180603
Subscription : [npglb2cms-1773207764316](#)
Resource group : [marchcohortbpaliwal](#)

Start time : 8/16/2026, 6:06:59 PM
Correlation ID : f07cce8b-651b-4e47-8dbe-ea9840c05d75

> Deployment details

✓ Next steps

Go to resource



Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)



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marchcohortbpaliwal

Resource group

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
- Help

Create

Manage view

Delete resource group

Refresh

Export to CSV

Open query

Assign tags

Move

Group by none

Essentials

JSON View

Resources Recommendations

Filter for any field...

Type equals all

Location equals all

Add filter

☐	Name ↑		Type	Location
☐	Application Insights Smart Detection	...	Action group	Global
☐	bsp-project2-resource	...	Foundry	East US 2
☐	bsp-project2-resource-appinsights	...	Application Insights	East US 2
☐	bsp-project2-resource-logs	...	Log Analytics workspace	East US 2
☐	bsp-project2 (bsp-project2-resource/bsp-project2)	...	Foundry project	East US 2
☐	bsp-project3-resource	...	Foundry	East US
☐	proj-wandernest (bsp-project3-resource/proj-wandernest)	...	Foundry project	East US
☐	bspprj3safety	...	Content safety	East US
☐	bspprj3speech	...	Speech service	East US
☐	bspprj3storage	...	Storage account	East US

Showing 1 - 10 of 11. Display count: auto

bspprj3safety

Content safety

Show platform metrics for this AI resource

Best practices for managing AI resources

Search

Delete

Overview

Essentials

JSON View

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Resource Management

Security

Monitoring

Automation

Help

Resource group (move) : [marchcohortbpaliwal](#)

Status : Active

Location : East US

Subscription (move) : [npglb2cms-1773207764316](#)

Subscription ID : a0e7e508-3870-4e84-b3d3-d1a183ee5089

Tags (edit) : [Add tags](#)

API Kind : ContentSafety

Pricing tier : Free

Endpoint : [https://bspprj3safety.cognitiveservices.azure.com/](#)

Manage keys : [Click here to manage keys](#)

Get Started

Monitoring

Congratulations! Your keys are ready.

Now explore the Quickstart guidance to get up and running with Content Safety.

1

Get the API Key to authenticate your applications and start sending calls to the service.

Every call to the Content Safety APIs requires the subscription key. The key can be found in the Keys and Endpoint section in the left pane. This key needs to be either passed through a query string parameter or specified in the request header.

2

Make an API call

Get in-depth information about the properties and methods of the API. Test your keys with the built-in testing console without writing a single line of code. Once you have the API running, you can check your consumption and the API health on Azure portal in your API 'Overview'.

Home > Storage center

Storage center | Blob Storage

GLB2CMS (npglb2cms.onmicrosoft.com)

Search

- Overview
- All storage resources
- Object storage
- Blob Storage**
- File storage
- Block storage
- Azure Disks
- Elastic SAN
- Data management
- Storage Discovery
- Storage Actions
- Migration
- Partner solutions
- Management services
- Help

Summary

Resources

+ Create

Restore

Manage view

Refresh

Export to CSV

Open query

Assign tags

Delete

Add to service group

Group by none

Create new storage account

Subscription equals all

Resource Group equals all

Location equals all

Add filter



No storage accounts to display

Create a storage account to store up to 500TB of data in the cloud. Use a general-purpose storage account to store object data, use a NoSQL data store, define and use queues for message processing, and set up file shares in the cloud. Use the Blob storage account and the hot or cool access tiers to optimize your costs based on how frequently your object data is accessed.

+ Create

[Learn more](#)

Showing 1 - 0 of 0. Display count: auto

[Give feedback](#)

Add or remove favorites by pressing Cmd+Shift+F

Create a storage account ...

×

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

npglb2cms-1773207764316

Resource group *

marchcohortbpaliwal

[Create new](#)

Instance details

Storage account name * ⓘ

bspprj3storage

Region * ⓘ

(US) East US

[Deploy to an Azure Extended Zone](#)

Primary service *

Azure Blob Storage or Azure Data Lake Storage

ⓘ This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

Performance * ⓘ

☒ **Standard:** Recommended for most scenarios (general-purpose v2 account)

☐ **Premium:** Recommended for scenarios that require low latency.

Redundancy * ⓘ

Locally redundant storage (LRS)

Create a storage account

- Basics
- Advanced
- Networking
- Data protection
- Security
- Encryption
- Tags
- Review + create

[View automation template](#)

Basics

Subscription	npglb2cms-1773207764316
Resource group	marchcohortbpaliwal
Location	East US
Storage account name	bspprj3storage
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled
Require Encryption in Transit for SMB	Enabled

Home

bspprj3storage_1786310597104

Deployment

Overview

Search

×

«

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : bspprj3storage_1786310597104
Subscription : [npglb2cms-1773207764316](#)
Resource group : [marchcohortbpaliwal](#)

Start time : 8/9/2026, 2:23:58 PM
Correlation ID : f5b0eebd-2f2b-4a37-9672-451781770523

Deployment details

Resource	Type	Status	Operation details
 bspprj3storage	 Storage account	Accepted	Operation details



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Home

bsprrj3storage_1786310597104 | Overview

Search

×

«

Delete Cancel Redeploy Download Refresh

- Overview
- Inputs
- Outputs
- Template

✓

Your deployment is complete

Deployment name : bsprrj3storage_1786310597104

Subscription : [npglb2cms-1773207764316](#)

Resource group : [marchcohortbpaliwal](#)

Start time : 8/9/2026, 2:24:01 PM

Correlation ID : f5b0eebd-2f2b-4a37-9672-451781770523

>

 Deployment details

✓

 Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

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Home > Storage center

Storage center | Blob Storage

GLB2CMS (npglb2cms.onmicrosoft.com)

Search

x

«

Summary

Resources

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All storage resources

Object storage

Blob Storage

File storage

Block storage

Azure Disks

Elastic SAN

Data management

Storage Discovery

Storage Actions

Migration

Partner solutions

Management services

Help

+ Create

↶ Restore

⚙ Manage view ▾

🔄 Refresh

↓ Export to CSV

🔗 Open query

🏷 Assign tags

🗑 Delete

+ Add to service group

☰ Group by none ▾

🔍 Filter for any field...

Subscription equals all

Resource Group equals all x

Location equals all x

+ Add filter



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+ Create

[Learn more](#)

Showing 1 - 0 of 0. Display count: auto ▾

Home > Storage center

Storage center | Blob Storage

GLB2CMS (npglb2cms.onmicrosoft.com)

Search

- Overview
- All storage resources
- Object storage
- Blob Storage**
- File storage
- Block storage
- Azure Disks
- Elastic SAN
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- Help

Summary

Resources

+ Create

Restore

Manage view

Refresh

Export to CSV

Open query

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Delete

Add to service group

Group by none

Create new storage account

Subscription equals all

Resource Group equals all

Location equals all

Add filter



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+ Create

[Learn more](#)

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Create a storage account ...

×

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Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

npglb2cms-1773207764316

Resource group *

marchcohortbpaliwal

[Create new](#)

Instance details

Storage account name * ⓘ

bspprj3storage

Region * ⓘ

(US) East US

[Deploy to an Azure Extended Zone](#)

Primary service *

Azure Blob Storage or Azure Data Lake Storage

ⓘ This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

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Redundancy * ⓘ

Locally redundant storage (LRS)

Create a storage account



- Basics
- Advanced
- Networking
- Data protection
- Security
- Encryption
- Tags
- Review + create

 [View automation template](#)

Basics

Subscription	npglb2cms-1773207764316
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Location	East US
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Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

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Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled
Require Encryption in Transit for SMB	Enabled

Previous

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Create

 Give feedback

Home

bspprj3storage_1786310597104 | Overview

Deployment



Search



Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete



Deployment name : bspprj3storage_1786310597104

Subscription : npglb2cms-1773207764316

Resource group : marchcohortbpaliwal

Start time : 8/9/2026, 2:24:01 PM

Correlation ID : f5b0eebd-2f2b-4a37-9672-451781770523

> Deployment details

✓ Next steps

Go to resource



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Azure services



Create a resource



Storage accounts



AI Search (Foundry IQ)



Foundry



Resource groups



Quickstart Center



Kubernetes services



Virtual machines



App Services



More services

Resources

Recent Favorite

Name	Type	Last Viewed
 bspprj3storage	Storage account	a few seconds ago
 bsp-project3-resource	Foundry	39 minutes ago
 bspproj3search	Search service (Foundry IQ)	39 minutes ago
 bsp-project2-resource	Foundry	6 days ago

See all

Navigate



Subscriptions



Resource groups



All resources



Dashboard

Tools

Azure AI Search Service Creation

Azure AI Search Resource creation

(Put the screenshots and details during Azure AI Search Resource creation here)

Azure services



Create a resource



Storage accounts



AI Search (Foundry IQ)



Foundry



Resource groups



Quickstart Center



Kubernetes services



Virtual machines



App Services



More services

Resources

Recent Favorite

Name	Type	Last Viewed
 bspprj3storage	Storage account	a few seconds ago
 bsp-project3-resource	Foundry	39 minutes ago
 bspproj3search	Search service (Foundry IQ)	39 minutes ago
 bsp-project2-resource	Foundry	6 days ago

See all

Navigate



Subscriptions



Resource groups



All resources



Dashboard

Tools

Home > Microsoft Foundry

Microsoft Foundry | AI Search

...

GLB2CMS

Search

x

<<

+ Create

⚙️ Manage view ▾

🔄 Refresh

⬇️ Export to CSV

🔗 Open query

|

🏷️ Assign tags

🗑️ Delete

+ Add to service group

📊 Group by none ▾

- 👤 Overview
- 📊 All resources
- ▾ Use with Foundry
- 👤 Foundry
- 👤 AI Hubs
- 🌟 Azure OpenAI
- ☁️ AI Search
- > More services
- > Classic AI services
- > Help

🔍 Filter for any field...

Subscription equals all

Resource Group equals all x

Location equals all x

+ Add filter



No search services to display

Information retrieval at scale for generative AI (RAG) and classic search over user-owned content.

+ Create

[Learn more](#)

Showing 1 - 0 of 0. Display count: auto ▾

[Give feedback](#)

Home > Microsoft Foundry

 Microsoft Foundry | AI Search ...

GLB2CMS

 Search

×

«

+ Create

 Manage view ▾

 Refresh

 Export to CSV

 Open query

|

 Assign tags

 Delete

+

Add to service group

 Group by none ▾

-  Overview
-  All resources
- ▾ Use with Foundry
-  Foundry
-  AI Hubs
-  Azure OpenAI
-  AI Search
- > More services
- > Classic AI services
- > Help

Create new search service ...

Subscription equals all

Resource Group equals all ×

Location equals all ×

+ Add filter



No search services to display

Information retrieval at scale for generative AI (RAG) and classic search over user-owned content.

+ Create

[Learn more](#)

Showing 1 - 0 of 0. Display count:

auto ▾

 Give feedback

Create a search service ...

- Basics
- Scale
- Tags
- Review + create

Project details

Subscription *

npglb2cms-1773207764316

Resource group *

marchcohortbpaliwal

Create new

Instance Details

Service name * ⓘ

prj3search

Location *

(US) East US 2 EUAP

You may be able to create a search service with higher storage limits in another region. View the full list of regions in the documentation. [Learn more](#)

Pricing tier * ⓘ

Free

50 MB, max 1 replicas, max 1 partitions, max 1 search units

[Change Pricing Tier](#)

Create a search service



- Basics
- Scale
- Tags
- Review + create

Basics

Subscription	npglb2cms-1773207764316
Resource Group	marchcohortbpaliwal
Location	East US 2 EUAP
Service name	prj3search
Pricing tier	Free (50 MB, max 1 replicas, max 1 partitions, max 1 search units)
Estimated cost per month	\$0.00
Compute type	Default

Scale

Replicas	1
Partitions	1

Networking

Endpoint connectivity (data)	Public
------------------------------	--------

Encryption

Encryption type	Microsoft-managed keys
Enforce CMK encryption	Disabled

Home

searchservice-1786165238377 | Overview

Deployment

Search

×

«

Delete

Cancel

Redeploy

Download

Refresh

- Overview
- Inputs
- Outputs
- Template

Deployment is in progress

Deployment name : searchservice-1786165238377

Subscription : npglb2cms-1773207764316

Resource group : marchcohortbpaliwal

Start time : 8/7/2026, 10:00:42 PM

Correlation ID : c6ae5a7d-b602-44b7-b816-69d2eb39934b

Deployment details

Resource	Type	Status	Operation details
There are no resources to display.			



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Home

 searchservice-1786165238377 | Overview ...

Deployment

Search

×

«

 Delete  Cancel  Redeploy  Download  Refresh

-  Overview
-  Inputs
-  Outputs
-  Template

 **Your deployment is complete**

 Deployment name : searchservice-1786165238377
Subscription : [npglb2cms-1773207764316](#)
Resource group : [marchcohortbpaliwal](#)

Start time : 8/7/2026, 10:00:42 PM
Correlation ID : c6ae5a7d-b602-44b7-b816-69d2eb39934b

 Deployment details

 Next steps

[Go to resource](#)

 **Cost management**
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

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prj3search

Search service (Foundry IQ)

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- > Agentic retrieval
- > Search management
- > Settings
- > Security + networking
- > Monitoring
- > Automation
- > Help

- + Add index
- Import data
- Search explorer
- Upgrade
- Refresh
- Delete
- Move

Essentials

JSON View

Resource group (move)	: marchcohortbpaliwal	Url	: https://prj3search.search.windows.net
Location (move)	: Australia East	Pricing tier	: Free
Subscription (move)	: npglb2cms-1773207764316	Replicas	: 1 (No SLA)
Subscription ID	: a0e7e508-3870-4e84-b3d3-d1a183ee5089	Partitions	: 1
Status	: Running	Search units	: 1
Date created	: Aug 7, 2026 at 10:01:01 PM	Compute type	: Default
Tags (edit)	: Add tags		

- Get started
- Properties
- Usage
- Monitoring

Revolutionary retrieval with Azure AI Search (Foundry IQ)

Don't know where to start? Here are some options from directly within the portal



Build your knowledge base



Connect your data



Monitor and scale

Azure Container Registry Service Creation

Azure Container Registry service creation

(Put the screenshots and details during Azure Container Registry Resource creation here)

Created the Container Registry as part of the ML Service
workspace creation

All services

Log Analytics workspaces

☆ ...

×

GLB2CMS (npglb2cms.onmicrosoft.com)

 Create  Open recycle bin  Manage view  Refresh  Export to CSV  Open query |  Assign tags + Add to service group

 Group by none

Create new log analytics workspace

Subscription equals all

Resource Group equals all

Location equals all

+ Add filter

☐	Name ↑		Resource Group	Location	Subscription
☐	 bsp-project2-resource-logs	...	marchcohortbpaliwal	East US 2	npglb2cms-1773207764316



Create Log Analytics workspace ...



Basics Tags Review + Create

i A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)



With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * **i**

npglb2cms-1773207764316



Resource group * **i**

marchcohortbpaliwal



[Create new](#)

Instance details

Name * **i**

bspproj3la



Region * **i**

East US



Review + Create

Review + Create

« Previous

Next : Tags >

All services > Log Analytics workspaces

Create Log Analytics workspace ...

Validation passed

Basics Tags Review + Create

 **Log Analytics workspace**
by Microsoft

Basics

Subscription	npglb2cms-1773207764316
Resource group	marchcohortbpaliwal
Name	bspproj3la
Region	East US

Pricing

Pricing tier	Pay-as-you-go (Per GB 2018)
--------------	-----------------------------

The cost of your workspace depends on the volume of data ingested and how long it is retained. Regional pricing details are available on the [Azure Monitor pricing page](#). You can change to a different pricing tier after the workspace is created. [Learn more](#) about Log Analytics pricing models.

Tags

None

All services

Microsoft.LogAnalyticsOMS | Overview

Deployment

Search

OverviewInputsOutputsTemplate

Delete

Cancel

Redeploy

Download

Refresh

Deployment is in progress

Deployment name : Microsoft.LogAnalyticsOMS

Subscription : [npglb2cms-1773207764316](#)

Resource group : [marchcohortbpaliwal](#)

Start time : 8/18/2026, 3:22:07 PM

Correlation ID : 65d91b9b-c7ba-4b7e-9618-f889726a5bbe

Deployment details

Resource	Type	Status	Operation details
bspproj3la	Log Analytics workspace	Created	Operation details



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All services



Microsoft.LogAnalyticsOMS | Overview

Deployment



Search



Delete



Cancel



Redeploy



Download



Refresh



Overview



Inputs



Outputs



Template



Your deployment is complete



Deployment name : Microsoft.LogAnalyticsOMS

Subscription : [npglb2cms-1773207764316](#)Resource group : [marchcohortbpaliwal](#)

Start time : 8/18/2026, 3:22:07 PM

Correlation ID : 65d91b9b-c7ba-4b7e-9618-f889726a5bbe



Deployment details



Next steps

[Go to resource](#)

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)

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Home

Application Insights

☆ ...

GLB2CMS (npglb2cms.onmicrosoft.com)

+ Create

⚙️ Manage view ▾

🔄 Refresh

⬇️ Export to CSV

🔗 Open query

|

🏷️ Assign tags

🗑️ Delete

+ Add to service group

☰ Group by none ▾

Create new Application Insights apps

Subscription equals all

Resource Group equals all ✕

Location equals all ✕

+ Add filter

☐	Name ↑		Resource Group	Location	Subscription
☐	💡 bsp-project2-resource-appinsights	...	marchcohortbpaliwal	East US 2	npglb2cms-1773207764316



Home > Application Insights

Application Insights

Monitor web app performance and usage

Create an Application Insights resource to monitor your live web application. With Application Insights, you have full observability into your application across all components and dependencies of your complex distributed architecture. It includes powerful analytics tools to help you diagnose issues and to understand what users actually do with your app. It's designed to help you continuously improve performance and usability. It works for apps on a wide variety of platforms including .NET, Node.js and Java EE, hosted on-premises, hybrid, or any public cloud. [Learn More](#)

PROJECT DETAILS

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

npglb2cms-1773207764316



Resource Group * ⓘ

marchcohortbpaliwal

[Create new](#)

INSTANCE DETAILS

Name * ⓘ

bspproj3appinsights



Region * ⓘ

(US) East US



OTLP support ⓘ

On

Off

WORKSPACE DETAILS

Subscription * ⓘ

npglb2cms-1773207764316



Log Analytics Workspace * ⓘ

(new) DefaultWorkspace-a0e7e508-3870-4e84-b3d3-d1a183ee5089-EUS ...

[Review + create](#)[« Previous](#)[Next : Tags >](#)

Application Insights

Monitor web app performance and usage

✔ Validation passed

- Basics
- Tags
- Review + create

SUMMARY

 **Application Insights**
by Microsoft

Subscription	npglb2cms-1773207764316
Resource Group	marchcohortbpaliwal
Name	bspproj3appinsights
Region	East US
Log Analytics Workspace	DefaultWorkspace-a0e7e508-3870-4e84-b3d3-d1a183ee5089-EUS [eastus]

Home

Microsoft.AppInsights | Overview

Deployment



Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Loading...



Deployment name : Microsoft.AppInsights
Subscription : [npglb2cms-1773207764316](#)
Resource group : [marchcohortbpaliwal](#)

Start time :
Correlation ID :

> Deployment details



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Home

Microsoft.AppInsights | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.AppInsights
Subscription : [npglb2cms-1773207764316](#)
Resource group : [marchcohortbpaliwal](#)

Start time : 8/18/2026, 3:26:59 PM
Correlation ID : c0b1c126-1a8b-4a62-98f4-25414581f1f2

> Deployment details

✓ Next steps

Go to resource



Cost management
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Home

Azure Machine Learning

☆ ...

×

GLB2CMS (npglb2cms.onmicrosoft.com)

+ Create ▾

Recently deleted

Manage view ▾

Refresh

Export to CSV

Open query

|

Assign tags

+ Add to service group

Group by none ▾

New workspace
For ML projects and teams

New registry
For sharing ML assets across workspaces

equals all

Type equals all

Resource Group equals all ×

Location equals all ×

+ Add filter



No workspaces to display

Workspaces are where you manage all the models, assets, and data related to your machine learning projects. Create one now to start using Azure Machine Learning.

[Learn more](#)



Home

Azure Machine Learning



GLB2CMS (npglb2cms.onmicrosoft.com)

+ Create ▾ Recently deleted Manage view ▾ Refresh Export to CSV Open query | Assign tags + Add to service group

Group by none ▾



Filter for any field...

Subscription equals all

Type equals all

Resource Group equals all



Location equals all



+ Add filter



No workspaces to display

Workspaces are where you manage all the models, assets, and data related to your machine learning projects. Create one now to start using Azure Machine Learning.

[Learn more](#)



Home > Azure Machine Learning

Azure Machine Learning

Create a machine learning workspace

Every workspace must be assigned to an Azure subscription, which is where billing happens. You use resource groups like folders to organize and manage resources, including the workspace you're about to create.

[Learn more about Azure resource groups](#)

Subscription * ⓘ

npglb2cms-1773207764316



Resource group * ⓘ

marchcohortbpaliwal

[Create new](#)

Workspace details

Configure your basic workspace settings like its storage connection, authentication, container, and more. [Learn more](#)

Name * ⓘ

bspproj3ml



Region * ⓘ

East US



Storage account * ⓘ

bspproj3storage

[Create new](#)

Key vault * ⓘ

(new) bspproj3ml4341922414

[Create new](#)

Application insights * ⓘ

bspproj3appinsights

[Create new](#)

Container registry ⓘ

None

[Create new](#)[Review + create](#)[< Previous](#)[Next : Inbound Access](#)

Create new container registry



Name * ⓘ

bspproj3containerregistry



SKU * ⓘ

Basic

[Save](#)[Discard](#)



Home > Azure Machine Learning

Azure Machine Learning



Create a machine learning workspace

Every workspace must be assigned to an Azure subscription, which is where billing happens. You use resource groups like folders to organize and manage resources, including the workspace you're about to create.

[Learn more about Azure resource groups](#)

Subscription *

①

npglb2cms-1773207764316

▼

Resource group *

①

marchcohortbpaliwal

▼

[Create new](#)

Workspace details

Configure your basic workspace settings like its storage connection, authentication, container, and more. [Learn more](#)

Name *

①

bspproj3ml

✓

Region *

①

East US

▼

Storage account *

①

bspproj3storage

▼

[Create new](#)

Key vault *

①

(new) bspproj3ml4341922414

▼

[Create new](#)

Application insights *

①

bspproj3appinsights

▼

[Create new](#)

Container registry

①

(new) bspproj3containerregistry

▼

[Create new](#)

[Review + create](#)[< Previous](#)[Next : Inbound Access](#)

Azure Machine Learning

...

Create a machine learning workspace

✕

✔ Validation passed

- Basics
- Inbound Access
- Outbound Access
- Encryption
- Identity
- Tags
- Review + create

Subscription	npglb2cms-1773207764316
Resource group	marchcohortbpaliwal
Region	East US
Name	bspproj3ml
Storage account	bsprrj3storage (marchcohortbpaliwal)
Key vault	(new) bspproj3ml4341922414
Application insights	bspproj3appinsights (marchcohortbpaliwal)
Container registry	(new) bspproj3containerregistry

Connectivity method	Enable public access from all networks
Network isolation	Public

Encryption type	Microsoft-managed keys
-----------------	------------------------

Identity type	System assigned
Enable HBI Flag	Disabled
Storage account access type	Credential-based
Shared key access	Enabled

Home

Microsoft.MachineLearningServices | Overview

Deployment

Search

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.MachineLearningServices
Subscription : [npglb2cms-1773207764316](#)
Resource group : [marchcohortbpaliwal](#)

Start time : 8/18/2026, 3:29:54 PM
Correlation ID : f2871221-8445-47a8-b3b2-96a994bc3c18

Deployment details

Resource	Type	Status	Operation details
bspproj3ml4341922414	Key vault	OK	Operation details



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Home

Microsoft.MachineLearningServices | Overview

Deployment

Search

Overview

Inputs

Outputs

Template

Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name : Microsoft.MachineLearningServices
Subscription : npglb2cms-1773207764316
Resource group : marchcohortbpaliwal

Start time : 8/18/2026, 3:29:55 PM
Correlation ID : f2871221-8445-47a8-b3b2-96a994bc3c18

Deployment details

Resource	Type	Status	Operation details
bspproj3ml	Microsoft.MachineLearningServi	OK	Operation details
bspproj3containerregistry	Container registry	OK	Operation details
bspproj3ml4341922414	Key vault	OK	Operation details

Next steps

Go to resource



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**bspproj3ml**

☆ ...

Azure Machine Learning workspace

Search

Download config.json

Delete

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Events
- Settings
- Monitoring
- Automation
- Support + troubleshooting

Essentials

Resource group	: marchcohortbpaliwal	Studio web URL	: https://ml.azure.com?tid=87a9af98-07c1-44f6-ae87-905a74...
Location	: East US	Container Registry	: bspproj3containerregistry
Subscription	: npglb2cms-1773207764316	Key Vault	: bspproj3ml4341922414
Storage	: bspproj3storage	Application Insights	: bspproj3appinsights
Provisioning State	: Succeeded	MLflow tracking URI	: azureml://eastus.api.azureml.ms/mlflow/v1.0/subscriptions/...

JSON View



Work with your models in Azure Machine Learning Studio

The Azure Machine Learning Studio is a web app where you can build, train, test, and deploy ML models. Launch it now to start exploring, or [learn more about the Azure Machine Learning studio](#)

Launch studio

Azure App Services

Azure App service creation

(Put the screenshots and details during Azure App Service creation here)

All services > App Services

Create Web App ...

×

- Basics
- Database
- Container
- Networking
- Monitor + secure
- Tags
- Review + create

Summary

 **Web App**
by Microsoft

Free sku
Estimated price - Free

 Basic authentication for this app is currently disabled and may impact deployments. Click to learn more.

Details

Subscription	a0e7e508-3870-4e84-b3d3-d1a183ee5089
Resource Group	marchcohortbpaliwal
Name	Wandernest
Secure unique default hostname	Enabled
Publish	Container
Site Container Name	main
Image:Tag	mcr.microsoft.com/appsvc/staticsite:latest
Server URL	https://mcr.microsoft.com
Port	80

App Service Plan (New)

Name: ASP-marchcohortbpaliwal-869F

Create Web App ...



Subscription * ⓘ

npglb2cms-1773207764316

Resource Group * ⓘ

marchcohortbpaliwal

Create new

Instance Details

Name

Wandernest

✓

-cmgvbsf8ftahc2f6.eastus-01.azurewebsites.net

Publish *

Code

Container

Operating System *

Linux

Windows

Region *

East US

ⓘ

 Not finding your App Service Plan? Try a different region or select your App Service Environment.

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app.
[Learn more](#)

Linux Plan (East US) * ⓘ

(New) ASP-marchcohortbpaliwal-9695

Create new

Pricing plan

Free F1 (1 GB memory)

Explore pricing plans

All services

App Services

GLB2CMS (npglb2cms.onmicrosoft.com)

+

Create

▼

🗑️

Manage Deleted Apps

⚙️

Manage view

▼

🔄

Refresh

⬇️

Export to CSV

🔗

Open query

🏷️

Assign tags

▶️

Start

↺️

Restart

⏏️

Stop

🗑️

Delete

⋮

📊

Group by none

▼

- + Web App
- + Web App + Database
- + WordPress on App Service
- + Web App (for Managed Instance)

Subscription equals all

Resource Group equals all

Type equals all

Location equals all

+ Add filter



No app services to display

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[Learn more about App Service](#)

Create Web App ...



Operation cannot be completed without additional quota.
Additional details - Location:
Current Limit (Total VMs): 0
Current Usage: 0
Amount required for this deployment (Total VMs): 1
(Minimum) New Limit that you should request to enable this deployment: 1.
Note that if you experience multiple scaling operations failing (in addition to this one) and need to accommodate the aggregate quota requirements of these operations, you will need to request a higher quota limit than the one currently displayed.

→

- Basics
- Database
- Container
- Networking
- Monitor + secure
- Tags
- Review + create

Summary

Web App
by Microsoft

Free sku

Estimated price - Free

Basic authentication for this app is currently disabled and may impact deployments. Click to learn more.

Details

Subscription	a0e7e508-3870-4e84-b3d3-d1a183ee5089
Resource Group	marchcohortbpaliwal
Name	Wandernest
Secure unique default hostname	Enabled
Publish	Container
Image:Tag	bspproj3containerregistry.azurecr.io/wandernest-ai:latest

Step 3: Project Architecture

Project Architecture: Folder Structure

(This section should include screenshots of the project folder structure and a brief description of each folder and its purpose.)

Folder	Purpose
src/	Core application implementation: Streamlit app, configuration, data loading, document ingestion, search engine, vector store, and evaluation.
data/	Source documents and the evaluation <code>golden_dataset.json</code> used by the application.
governance/	Governance controls including compliance checking, governance gating, and safety validation.
guardrails/	Runtime safety controls including Content Safety and PII detection.
tests/	Automated tests covering configuration, data loading, ingestion, search, vector store, safety, governance, Azure integration, and Streamlit functionality.
evidence/	Evidence artifacts documenting testing and validation of the implemented components and Azure integrations.
reports/	Evaluation outputs, including detailed evaluation results and the evaluation summary.
scripts/	Utility scripts; currently includes configuration checking.

```
bharat@macbookpro AIWithMicrosoftAzure % find Project-3 -maxdepth 2 -type f | sort
Project-3/.gitignore
Project-3/data/ai-schedule-change-policy-updated.pdf
Project-3/data/air-india-coc.pdf
Project-3/data/air-india-general-booking-policies-oct2025.pdf
Project-3/data/golden_dataset.json
Project-3/data/U.S. Department of Transportation - Air Consumer Privacy.pdf
Project-3/data/U.S. Department of Transportation - Aircraft Dissinjection.pdf
Project-3/data/U.S. Department of Transportation - Aviation Industry Bankruptcy and Service Cessation.pdf
Project-3/data/U.S. Department of Transportation - Implementation of the Consumer Credit Protection Act With Respect to Air Carriers and Foreign Air Carriers.pdf
Project-3/data/U.S. Department of Transportation - Refunds and Other Consumer Protections.pdf
Project-3/data/U.S. Department of Transportation - Refunds for Airline Fare and Ancillary Service Fees.pdf
Project-3/debug_vector_store.py
Project-3/Dockerfile
Project-3/evidence_environment_packages.txt
Project-3/evidence/evidence_all_tests.txt
Project-3/evidence/evidence_azure_openai_embeddings.txt
Project-3/evidence/evidence_azure_production_smoke.txt
Project-3/evidence/evidence_azure_search_index_sources.txt
Project-3/evidence/evidence_blob_azure_rag_smoke.txt
Project-3/evidence/evidence_chunking.txt
Project-3/evidence/evidence_content_safety.txt
Project-3/evidence/evidence_data_files.txt
Project-3/evidence/evidence_data_loader_blob.txt
Project-3/evidence/evidence_data_loader_local.txt
Project-3/evidence/evidence_data_loader_runtime.txt
Project-3/evidence/evidence_data_loader_tests.txt
Project-3/evidence/evidence_data_loader.txt
Project-3/evidence/evidence_document_categorization.txt
Project-3/evidence/evidence_document_ingestion_real_pipeline.txt
Project-3/evidence/evidence_document_ingestion_real.txt
Project-3/evidence/evidence_environment_config_tests.txt
Project-3/evidence/evidence_environment_package_imports.txt
Project-3/evidence/evidence_full_test_suite.txt
Project-3/evidence/evidence_governance_gate.txt
Project-3/evidence/evidence_loaded_document_metadata.txt
Project-3/evidence/evidence_missing_azure_document.txt
Project-3/evidence/evidence_safety_validator_azure.txt
Project-3/evidence/evidence_search_by_text_real.txt
Project-3/evidence/evidence_search_engine_initialization.txt
Project-3/evidence/evidence_search_engine_real_initialization.txt
Project-3/evidence/evidence_streamlit_search_real.txt
Project-3/evidence/evidence_streamlit_ui.txt
Project-3/evidence/evidence_synthesize_response_real.txt
Project-3/evidence/evidence_vector_store_integration.txt
Project-3/governance/__init__.py
Project-3/governance/compliance_checker.py
Project-3/governance/governance_gate.py
Project-3/governance/safety_validator.py
Project-3/guardrails/__init__.py
```

```

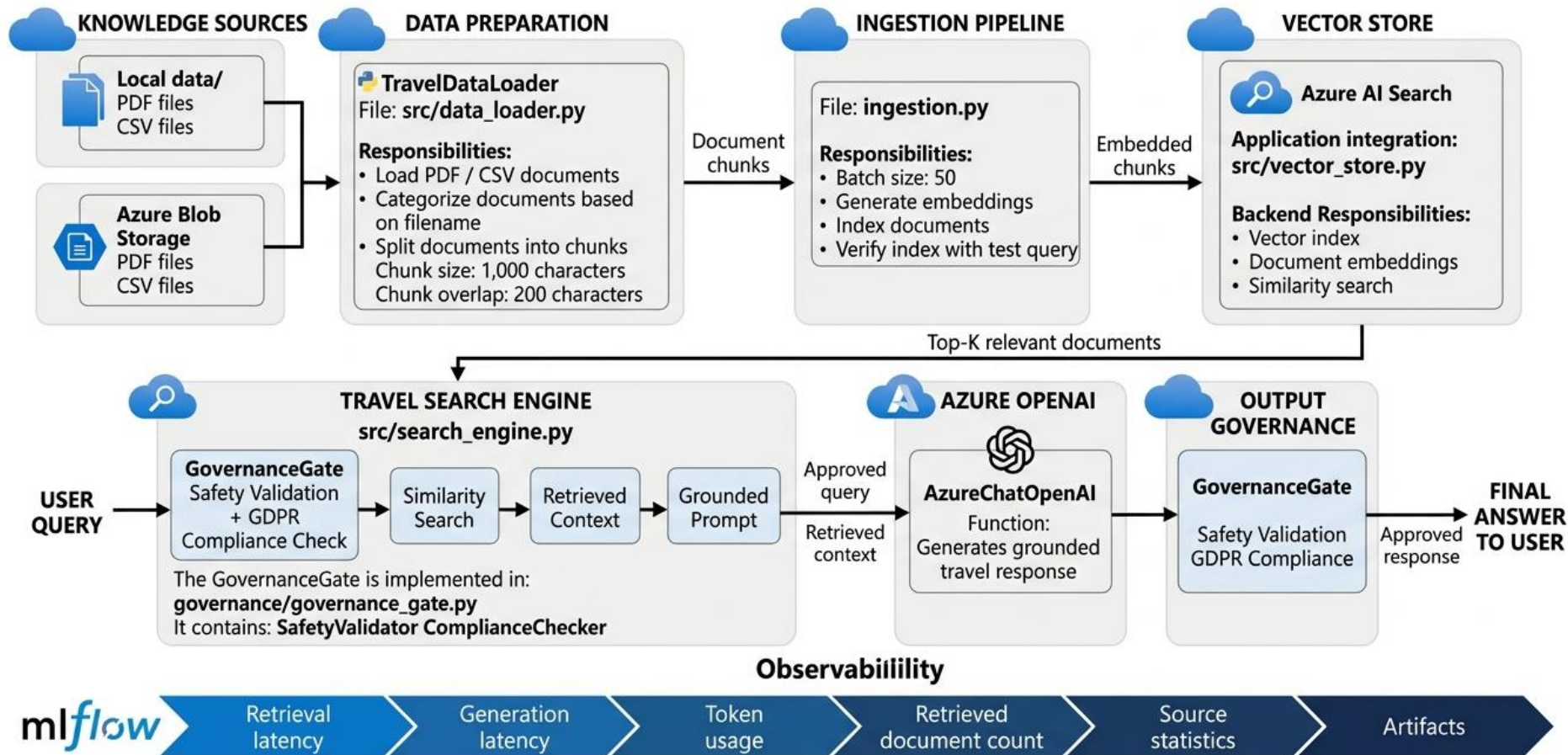
Project-3/evidence/evidence_vector_store_integration.txt
Project-3/governance/__init__.py
Project-3/governance/compliance_checker.py
Project-3/governance/governance_gate.py
Project-3/governance/safety_validator.py
Project-3/guardrails/__init__.py
Project-3/guardrails/content_safety.py
Project-3/guardrails/pii_detector.py
Project-3/README.md
Project-3/reports/evaluation_detailed.csv
Project-3/reports/evaluation_summary.json
Project-3/requirements.txt
Project-3/restart_mlflow.sh
Project-3/Rubric_report
Project-3/scripts/check_config.py
Project-3/src/app.py
Project-3/src/config.py
Project-3/src/data_loader.py
Project-3/src/evaluate.py
Project-3/src/ingestion.py
Project-3/src/search_engine.py
Project-3/src/vector_store.py
Project-3/standalone_embeddings.py
Project-3/standalone_vector_store.py
Project-3/start_mlflow.sh
Project-3/tests/__init__.py
Project-3/tests/test_azure_prod_smoke.py
Project-3/tests/test_azure_search_index_sources.py
Project-3/tests/test_blob_rag_smoke.py
Project-3/tests/test_config.py
Project-3/tests/test_content_safety.py
Project-3/tests/test_data_loader.py
Project-3/tests/test_document_ingestion_real_pipeline.py
Project-3/tests/test_document_ingestion_real.py
Project-3/tests/test_env_loading.py
Project-3/tests/test_governance_gate_real.py
Project-3/tests/test_missing_azure_document.py
Project-3/tests/test_safety_validator_azure_real.py
Project-3/tests/test_safety_validator.py
Project-3/tests/test_search_engine_initialization.py
Project-3/tests/test_search_engine_real_search.py
Project-3/tests/test_search_engine_real_synthesis.py
Project-3/tests/test_search_engine_real.py
Project-3/tests/test_search_engine.py
Project-3/tests/test_search.py
Project-3/tests/test_streamlit_search_real.py
Project-3/tests/test_streamlit_ui_real.py
Project-3/tests/test_vector_store_integration.py
Project-3/tests/test_vector_store.py
bharat@macbookpro AIWithMicrosoftAzure %

```

Project Architecture: Component Interaction

(This section should include a diagram and an explanation of how the components within the project folder structure interact with each other.)

Travel RAG System — End-to-End Component Architecture



Configure '.env'

(This section should include a screenshot of the credentials that need to be configured in the `.env` file, along with a brief description of each.)

Configuration

Purpose

AZURE_OPENAI_ENDPOINT	Azure OpenAI resource endpoint used for LLM inference and embeddings
AZURE_OPENAI_API_VERSION	Azure OpenAI API version
AZURE_OPENAI_DEPLOYMENT_NAME	GPT-5.4-mini deployment used for response generation
AZURE_OPENAI_EMBEDDING_DEPLOYMENT	<code>text-embedding-3-small</code> deployment used to generate document/query embeddings
AZURE_OPENAI_API_KEY	Authentication credential for Azure OpenAI
AZURE_SEARCH_ENDPOINT	Azure AI Search endpoint hosting the vector index
AZURE_SEARCH_KEY	Authentication credential for Azure AI Search
AZURE_SEARCH_INDEX_NAME	<code>bspproject3index</code> — vector/search index used for RAG retrieval
VECTOR_STORE_TYPE_PROD	Production vector store selection: <code>azure_search</code>
AZURE_CONTENT_SAFETY_ENDPOINT	Azure AI Content Safety endpoint
AZURE_CONTENT_SAFETY_KEY	Authentication credential for Content Safety

Configuration

Purpose

MLFLOW_EXPERIMENT_NAME

`travel-search-rag` — experiment used to track RAG runs, parameters and metrics

AZURE_STORAGE_CONNECTION_STRING

Authentication to Azure Blob Storage

AZURE_STORAGE_CONTAINER_NAME

`project3` — container holding the source documents

BLOB_STORAGE

`True` — application loads documents from Azure Blob Storage

AZURE_TENANT_ID

Azure Entra ID tenant used for authentication

AZURE_SUBSCRIPTION_ID

Azure subscription containing the project resources

AZURE_IDENTITY_DISABLE_INTERACTIVE_BROWSER_CREDENTIAL

`true` — prevents interactive browser authentication in the container

GIT_PYTHON_REFRESH

`quiet` — suppresses non-critical Git/MLflow environment warnings

.env

- Azure OpenAI configuration
- Azure AI Search configuration
- Azure Content Safety configuration
- Azure Blob Storage configuration
- Azure ML / MLflow configuration
- Azure identity configuration
 - AZURE_IDENTITY_DISABLE_INTERACTIVE_BROWSER_CREDENTIAL=true

~/.azure

- azureProfile.json
- msal_token_cache.json [REDACTED]
- clouds.config
- Azure CLI authentication/session state

Authentication: Azure CLI credentials are persisted in the container's `~/.azure` directory. The application uses non-interactive Azure identity authentication rather than prompting the user for browser/device-code authentication during application execution.

Roadmap - once Azure App Quota is resolved

Azure Deployment

For local Docker execution, Azure CLI authentication is provided through `~/ .azure`, while service configuration is supplied through `.env`.

For Azure-hosted deployment, the application will use **Managed Identity / Microsoft Entra ID** instead of personal Azure CLI credentials. Required RBAC permissions will be assigned to the application's managed identity for Azure ML, Blob Storage, Azure AI Search, and other Azure resources. Secrets will be stored in **Azure Key Vault** rather than embedded in the application configuration.

Step 4: Running the Application

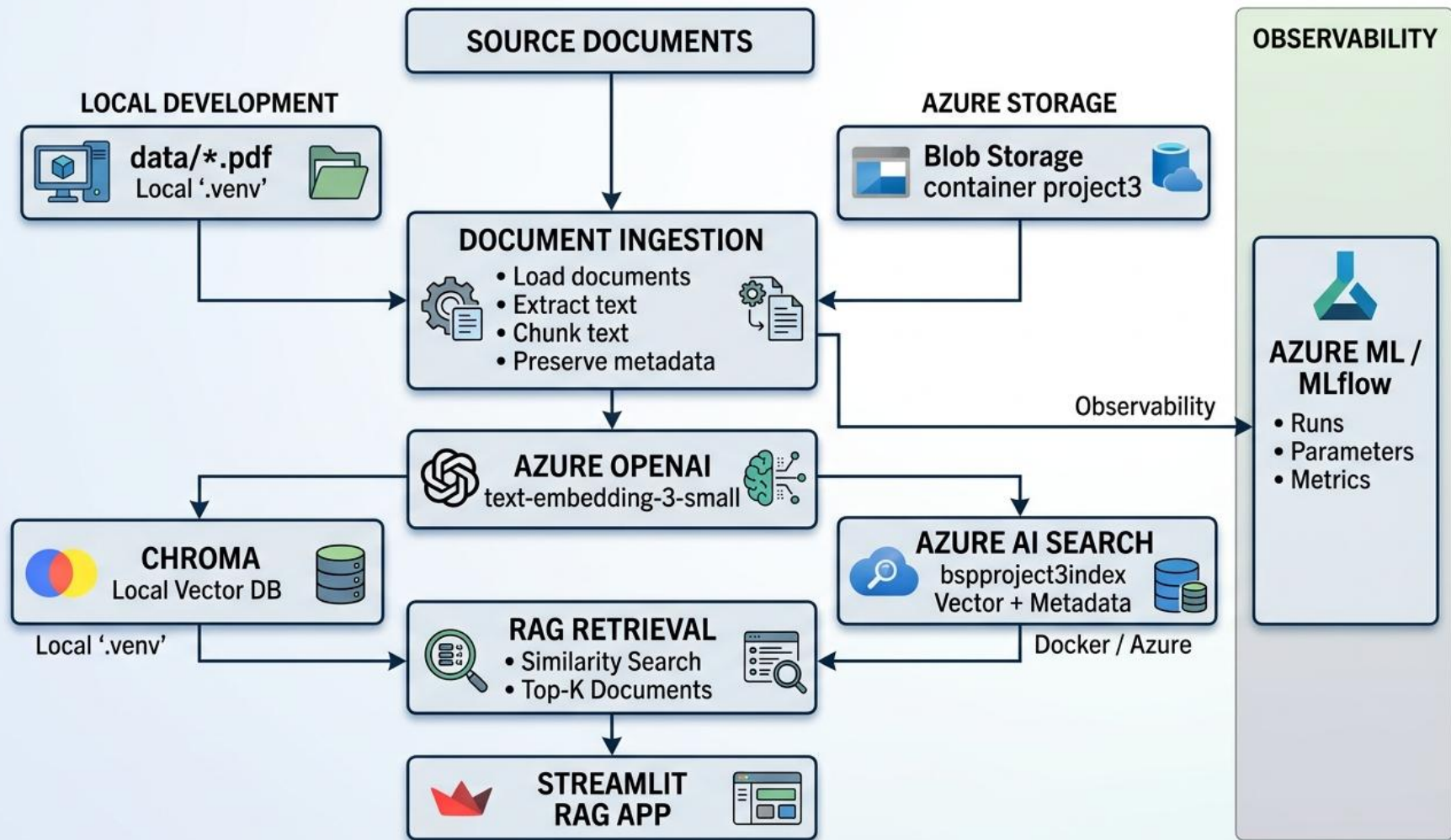
(Data Ingestion → Application → Evaluation)

4.1 Data Ingestion Execution

(This section should describe how to execute the data ingestion process and include a screenshot of the output)

The ingestion pipeline was implemented and validated across three environments:

Environment	Storage	Vector Store	Purpose
Local .venv	Local data/ directory	Chroma	Initial development and testing
Local Docker	Azure Blob Storage	Azure AI Search	Validate production-like containerized execution
Azure Deployment	Azure Blob Storage	Azure AI Search	Production architecture



Data Ingestion Pipeline — Azure Production Flow

Source

- 9 travel-policy PDFs stored in **Azure Blob Storage**
- Blob container: `project3`
- `BLOB_STORAGE=True` enables cloud-based document loading

Processing

- `TravelDataLoader` loads documents from Azure Blob Storage
- 152 document pages loaded
- Documents split into **1,018 chunks**
- Average chunk size: **919 characters**

Embedding & Indexing

- Azure OpenAI `text-embedding-3-small`
- LangChain `AzureSearch` vector store
- Azure AI Search index: `bspproject3index`
- Batch size: **50 chunks**

Verification

- Similarity search executed with `k=5`
- **5 documents retrieved**
- Index verification successful

python3 -m src.ingestion

root@7a78d8acce5f:/app# python3 -m src.ingestion

DEBUG: Running in Docker/container environment. Using environment variables; .env is not required.

 Starting Travel Document Ingestion

=====

 Azure Blob Storage enabled: container='project3'

DEBUG: MLflow configured successfully.

DEBUG: Tracking URI:

azureml://eastus.api.azureml.ms/mlflow/v1.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml

DEBUG: Azure Content Safety client initialized successfully.

DEBUG: Initializing AzureChatOpenAI with endpoint https://bsp-project3-resource.openai.azure.com and deployment gpt-5.4-mini

DEBUG: AzureChatOpenAI initialized successfully.

DEBUG: Initializing AzureOpenAIEmbeddings with endpoint https://bsp-project3-resource.openai.azure.com and deployment text-embedding-3-small

DEBUG: AzureOpenAIEmbeddings initialized successfully.

DEBUG: Initializing vector store...

DEBUG: Configured Azure Search endpoint: https://bspproject3.search.windows.net

DEBUG: Configured Azure Search index: bspproject3index

DEBUG: Initializing AzureSearch vector store...

DEBUG: AzureSearch vector store initialized successfully.

 Loading travel knowledge base...

DEBUG: Vector store initialized successfully.

python3 -m src.ingestion

Source: Azure Blob Storage

=====

☁ Loading PDFs from Azure Blob container 'project3'...

Found 9 PDF files in Azure Blob Storage

Downloading: U.S. Department of Transportation - Air Consumer Privacy.pdf

✓ Loaded: U.S. Department of Transportation - Air Consumer Privacy.pdf (3 pages)

Downloading: U.S. Department of Transportation - Aircraft Dissinsection.pdf

✓ Loaded: U.S. Department of Transportation - Aircraft Dissinsection.pdf (5 pages)

Downloading: U.S. Department of Transportation - Aviation Industry Bankruptcy and Service Cessation.pdf

✓ Loaded: U.S. Department of Transportation - Aviation Industry Bankruptcy and Service Cessation.pdf (5 pages)

Downloading: U.S. Department of Transportation - Implementation of the Consumer Credit Protection Act With Respect to Air Carriers and Foreign Air Carriers.pdf

✓ Loaded: U.S. Department of Transportation - Implementation of the Consumer Credit Protection Act With Respect to Air Carriers and Foreign Air Carriers.pdf (2 pages)

Downloading: U.S. Department of Transportation - Refunds and Other Consumer Protections.pdf

✓ Loaded: U.S. Department of Transportation - Refunds and Other Consumer Protections.pdf (80 pages)

Downloading: U.S. Department of Transportation - Refunds for Airline Fare and Ancillary Service Fees.pdf

✓ Loaded: U.S. Department of Transportation - Refunds for Airline Fare and Ancillary Service Fees.pdf (9 pages)

Downloading: ai-schedule-change-policy-updated.pdf

✓ Loaded: ai-schedule-change-policy-updated.pdf (1 pages)

Downloading: air-india-coc.pdf

✓ Loaded: air-india-coc.pdf (28 pages)

Found 0 CSV files in Azure Blob Storage

DEBUG: Similarity search returned 5 documents.

python3 -m src.ingestion

DEBUG: Retrieval latency: 0.897s

DEBUG: Unique source documents: 2

✅ Index verification successful!

Test query: 'What are the baggage allowance rules for international flights?'

Retrieved: 5 documents

 View run document_ingestion at:

<https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5/runs/4d2bce7c-7187-44b6-bbc8-0046db27ad46>

 View experiment at:

<https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5>

=====

 Ingestion pipeline completed!

root@7a78d8acce5f:/app#

=====

Found 9 PDF files in Azure Blob Storage

✓ Loaded: U.S. Department of Transportation - Air Consumer Privacy.pdf (3

✓ Loaded: U.S. Department of Transportation - Aircraft Dissinsection.pdf (5 pages)

✓ Loaded: U.S. Department of Transportation - Aviation Industry Bankruptcy and Service Cessation.pdf (5

Downloaded: 03/03/2016 Department of Transportation - Implementation of the Consumer Credit Protection Act With Respect to Air Carriers and Foreign Air Carriers.pdf (2)

Downloaded: U.S. Department of Transportation - Refunds and Other Consumer Protections.pdf
 < Loaded: U.S. Department of Transportation - Refunds and Other Consumer Protections.pdf (8

Downloaded: U.S. Department of Transportation - Refunds for Airline Fare and Ancillary Service Fees.pdf
 : Loaded: U.S. Department of Transportation - Refunds for Airline Fare and Ancillary Service Fees.pdf (0

Downloading: ai-schedule-change-policy-updated.pdf

Downloading: air-india-coc.pdf

Downloading: air-india-general-booking-policies-oct2025.pdf

Looking for CSV files in Azure Blob container 'project3'...

.....

✓ Created 1018 chunks

```
Total chunks to index: 1018
```

Indexing documents to Azure AI Search

indexing: 100% | ██████████ 21/21 [00:49:00.00 / 2:18:14]

✓ Ingestion complete:
Successfully indexed: 1018 chunks

Verifying index

DEBUG: Text Query: What are the baggage allowance rules for international flights?

```
ll start_run with nested=true
=====
```

```
DEBUG: Similarity search returned 5 documents.
```

```
DEBUG: Unique source documents: 2
```

Test query: 'What are the baggage allowance rules for international flights?'

[🏃 View run document_inge](#)

View experiment at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/res>

 Ingestion pipeline completed!

```
root@7a78d8acce5f:/app#
```

4.2 Running the Streamlit Application

(This section should describe how to execute the Streamlit application and include a screenshot of the output UI)

Running the Streamlit Application

Application: WanderNest Travel RAG Chatbot

Framework: Streamlit

Entry Point: [src/app.py](#)

Local execution

```
source .venv/bin/activate  
streamlit run src/app.py
```

Docker execution

```
docker build -t project3-rag .  
docker run --rm -it -p 8501:8501 \  
-v "$HOME/.azure:/root/.azure" --env-file .env \  
project3-rag streamlit run src/app.py \  
--server.address=0.0.0.0
```

The container runs:

streamlit run src/app.py

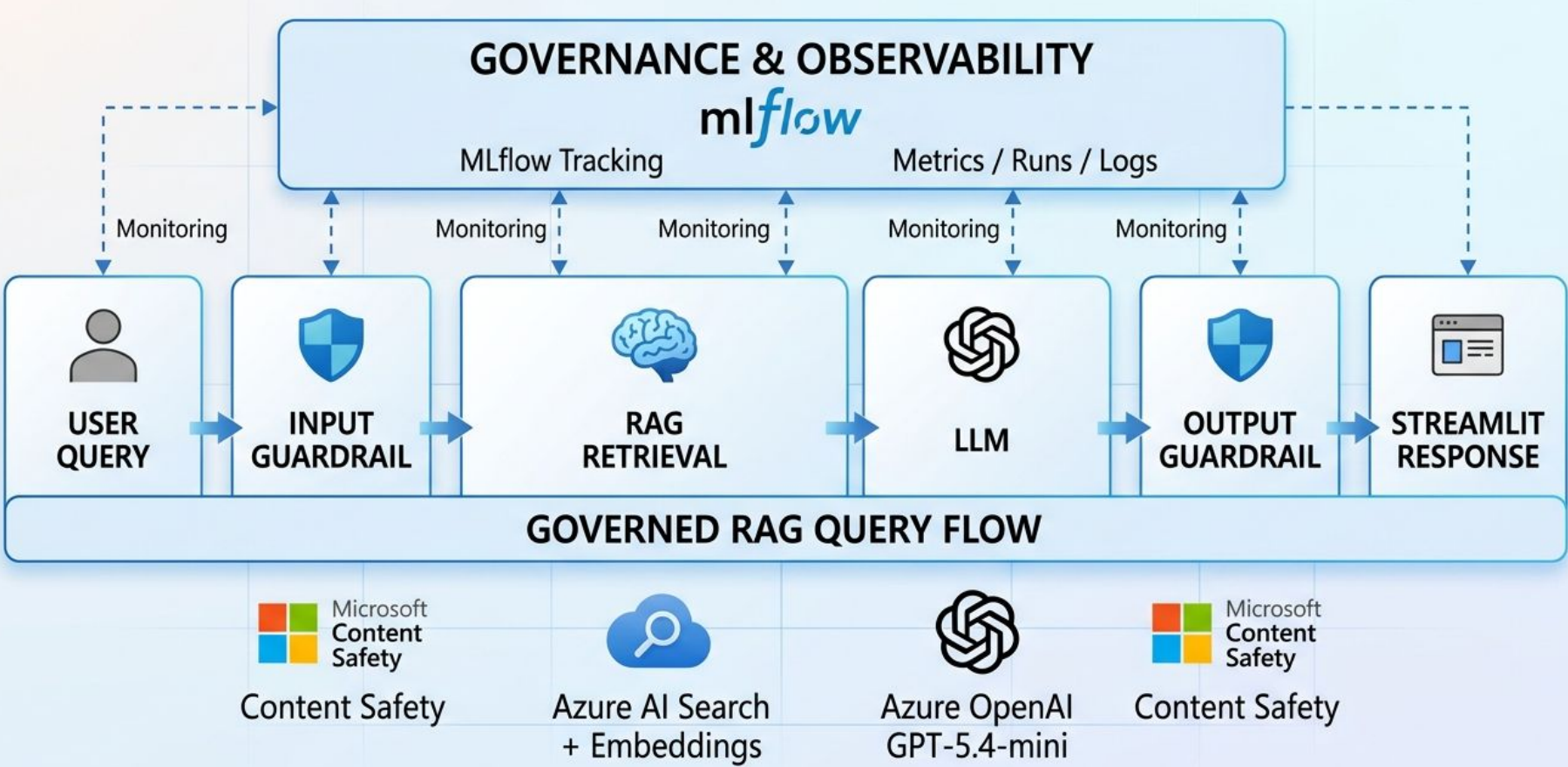
Current Docker deployment

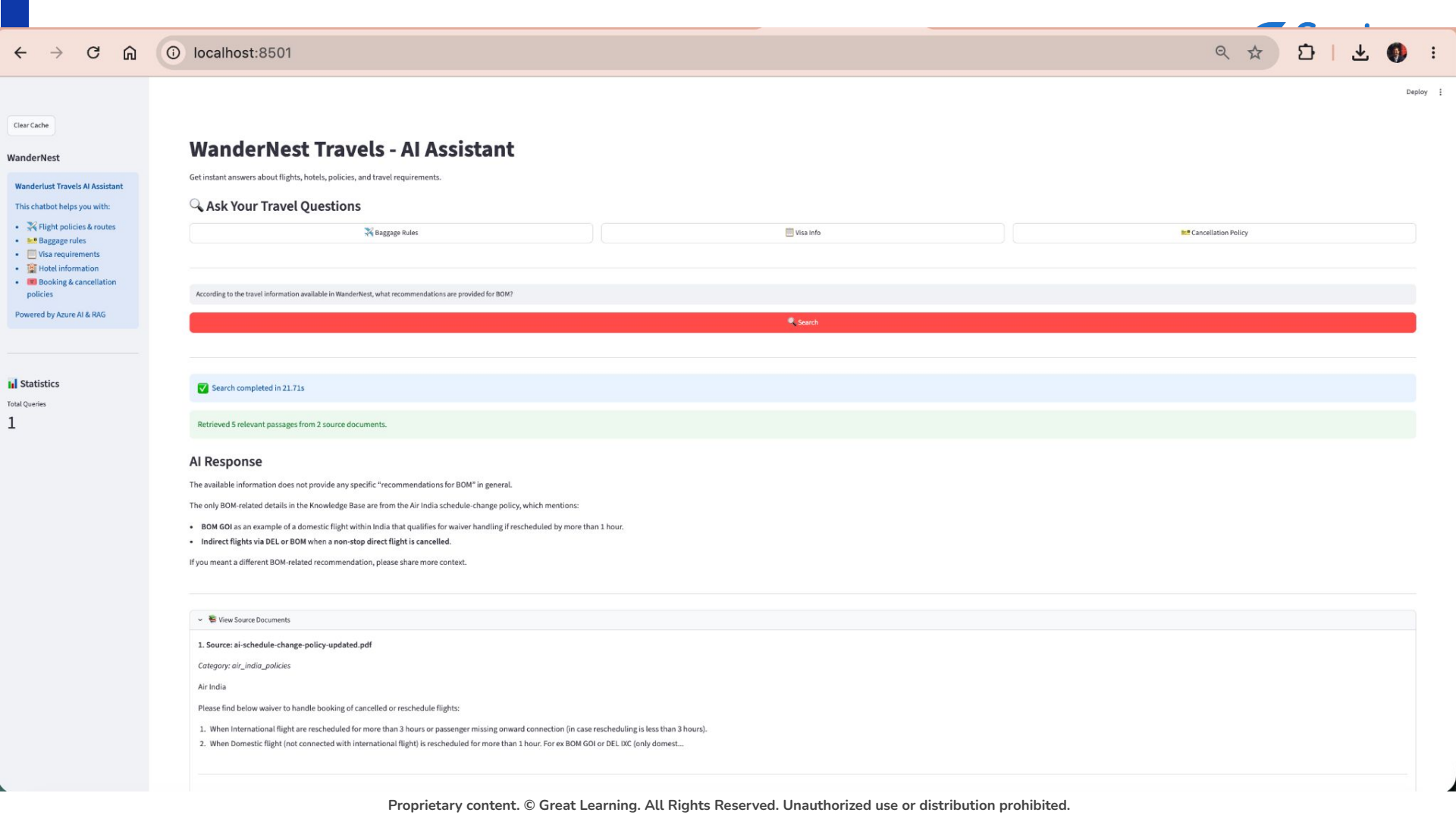
- Containerized application: [project3-rag](#)
- Port mapping: [8501:8501](#)
- Configuration supplied through environment variables
- Azure services used as production backends

Key point:

The same application code can run in the local Python environment and inside Docker, while the Docker configuration connects to the Azure production services.

WanderNest GenAI Travel Chatbot - Governed RAG Query Flow





WanderNest Travels - AI Assistant

Get instant answers about flights, hotels, policies, and travel requirements.

Ask Your Travel Questions

Baggage Rules

Visa Info

Cancellation Policy

According to the travel information available in WanderNest, what recommendations are provided for BOM?

Search

✓ Search completed in 21.71s

Retrieved 5 relevant passages from 2 source documents.

AI Response

The available information does not provide any specific "recommendations for BOM" in general.

The only BOM-related details in the Knowledge Base are from the Air India schedule-change policy, which mentions:

- **BOM GOI** as an example of a domestic flight within India that qualifies for waiver handling if rescheduled by more than 1 hour.
- **Indirect flights via DEL or BOM** when a **non-stop direct flight** is cancelled.

If you meant a different BOM-related recommendation, please share more context.

View Source Documents

1. Source: ai-schedule-change-policy-updated.pdf

Category: air_india_policies

Air India

Please find below waiver to handle booking of cancelled or reschedule flights:

1. When International flight are rescheduled for more than 3 hours or passenger missing onward connection (in case rescheduling is less than 3 hours).
2. When Domestic flight (not connected with international flight) is rescheduled for more than 1 hour. For ex BOM GOI or DEL IXC (only domest...

DEBUG: Starting answer_query()
DEBUG: Starting retrieval...
DEBUG: Text Query: According to the travel information available in WanderNest, what recommendations are provided for BOM?
DEBUG: Mlflow run started: search_by_text
DEBUG: Performing similarity search with k=5
DEBUG: Similarity search returned 5 documents.
DEBUG: Retrieval latency: 1.118s
DEBUG: Unique source documents: 2

✖ View run search_by_text at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5/runs/4918d96e-acd3-4818-9731-7c90555c9932>

✔ View experiment at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5>

DEBUG: Mlflow run completed: search_by_text

DEBUG: Retrieval completed in 9.267s

DEBUG: Documents found: 5

DEBUG: Unique sources: 2

DEBUG: Starting response generation...

DEBUG: Mlflow run started: synthesize_response

DEBUG: Generating response using Azure OpenAI...

DEBUG: LLM response generated successfully.

DEBUG: LLM generation latency: 1.886s

DEBUG: Token usage - input=1385, output=115, total=1500

✖ View run synthesize_response at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5/runs/a57a1aaf-3b25-4c08-9c89-30bb46c5e492>

✔ View experiment at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5>

DEBUG: Mlflow run completed: synthesize_response

DEBUG: Generation completed successfully.

DEBUG: Generation latency: 1.886s

DEBUG: Token usage - input=1385, output=115, total=1500

DEBUG: answer_query() completed successfully.

DEBUG: Starting answer_query()

DEBUG: Starting retrieval...

DEBUG: Text Query: According to the travel information available in WanderNest, what recommendations are provided for BOM?

DEBUG: Mlflow run started: search_by_text

DEBUG: Performing similarity search with k=5

DEBUG: Similarity search returned 5 documents.

DEBUG: Retrieval latency: 0.789s

DEBUG: Unique source documents: 2

✖ View run search_by_text at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5/runs/b8811eb5-9db0-499e-9775-7416e279b10c>

✔ View experiment at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5>

DEBUG: Mlflow run completed: search_by_text

DEBUG: Retrieval completed in 8.827s

DEBUG: Documents found: 5

DEBUG: Unique sources: 2

DEBUG: Starting response generation...

DEBUG: Mlflow run started: synthesize_response

DEBUG: Generating response using Azure OpenAI...

DEBUG: LLM response generated successfully.

DEBUG: LLM generation latency: 1.780s

DEBUG: Token usage - input=1385, output=107, total=1492

✖ View run synthesize_response at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5/runs/afc8c6e1-2760-445f-947a-8040ea7d7d0c>

✔ View experiment at: <https://eastus.api.azureml.ms/mlflow/v2.0/subscriptions/a0e7e508-3870-4e84-b3d3-d1a183ee5089/resourceGroups/marchcohortbpaliwal/providers/Microsoft.MachineLearningServices/workspaces/bspproj3ml/#/experiments/a4d5040c-45ff-48a5-bb11-0c87edef00b5>

DEBUG: Mlflow run completed: synthesize_response

DEBUG: Generation completed successfully.

DEBUG: Generation latency: 1.780s

DEBUG: Token usage - input=1385, output=107, total=1492

DEBUG: answer_query() completed successfully.

4.2 Running RAGAS Evaluation

(This section should describe how to run a RAGAS evaluation to assess the performance of the RAG system, explain the evaluation metrics, and include a screenshot of the output)

RAGAS Evaluation Metrics

Metric	What it measures	Desired behavior
Faithfulness	Whether the generated answer is supported by the retrieved context	Higher is better
Answer Relevancy	Whether the generated answer directly addresses the user's question	Higher is better
Context Precision	Whether the retrieved context contains relevant information, with relevant chunks ranked appropriately	Higher is better
Context Recall	Whether the retrieved context contains the information needed to answer the question	Higher is better

RAGAS EVALUATION

RETRIEVAL QUALITY



**Context
Precision**



**Context
Recall**

**Did we retrieve the
right context?**

ANSWER QUALITY



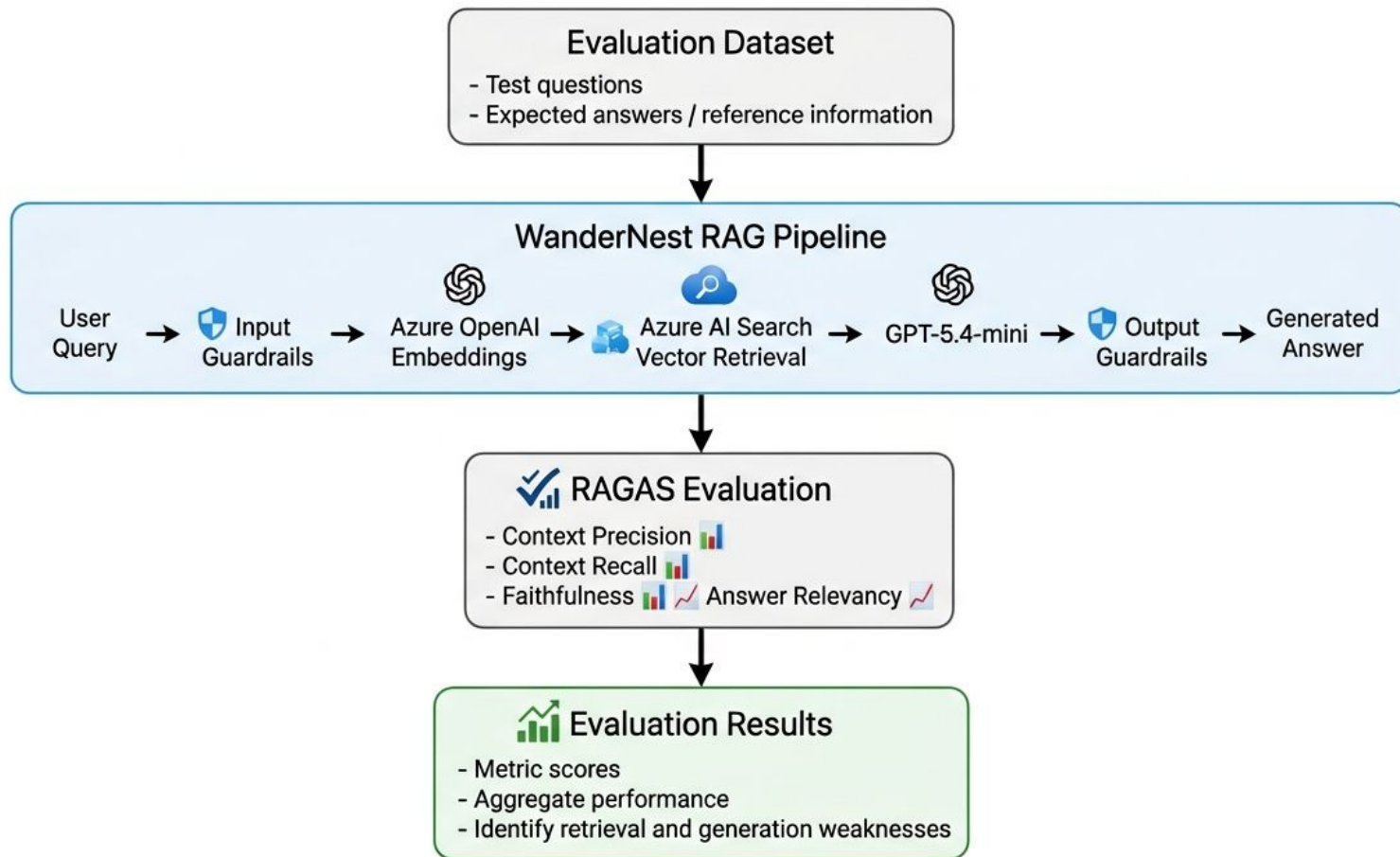
Faithfulness



**Answer
Relevancy**

**Did we generate the
right answer?**

WanderNest — RAGAS Evaluation Pipeline



Evaluation reports for improving metrics

Metric	750 / 150	1000 / 200 - Selected
Context Precision	0.8555	0.9079
Context Recall	0.8000	1.0000
Faithfulness	0.7228	0.8089
Answer Relevancy	0.7811	0.7739
Evidence	evidence/ ├── evidence_ingestion_750_150.txt ├── └── evidence_evaluation_750_150.txt	evidence/ ├── evidence_ingestion_1000_200.txt ├── evidence_evaluation_1000_200.txt



```
[root@0b4808988c7e:/app# grep -nEi "Context Precision:|Context Recall:|Faithfulne
ss:|Answer Relevancy:|EVALUATION PASSED|EVALUATION FAILED" evidence/evidence_eva
luation_1000_200.txt
```

```
2476:INFO:evaluation: Faithfulness: 0.8089
```

```
2477:INFO:evaluation: Answer Relevancy: 0.7739
```

```
2478:INFO:evaluation: Context Precision: 0.9079
```

```
2479:INFO:evaluation: Context Recall: 1.0000
```

```
2490:INFO:evaluation: ✓ EVALUATION PASSED
```

```
2653:INFO:evaluation: ✓ EVALUATION PASSED
```

```
[root@0b4808988c7e:/app# grep -nEi "Context Precision:|Context Recall:|Faithfulne
ss:|Answer Relevancy:|EVALUATION PASSED|EVALUATION FAILED" evidence/evidence_eva
luation_750_150.txt
```

```
2475:INFO:evaluation: Faithfulness: 0.7228
```

```
2476:INFO:evaluation: Answer Relevancy: 0.7811
```

```
2477:INFO:evaluation: Context Precision: 0.8555
```

```
2478:INFO:evaluation: Context Recall: 0.8000
```

```
2489:INFO:evaluation: ✓ EVALUATION PASSED
```

```
2652:INFO:evaluation: ✓ EVALUATION PASSED
```

```
root@0b4808988c7e:/app#
```


Conclusion : Key Takeaways and Business Outcomes

Key Takeaways

- **End-to-end GenAI solution on Microsoft Azure** — from document ingestion and storage to RAG retrieval, response generation, governance, evaluation, and monitoring. *(not deployed due to resource quota issues)*
- **Enterprise-ready RAG architecture** combines Azure Blob Storage, Azure AI Search, Azure OpenAI, Azure Content Safety, and Azure ML/MLflow.
- **Grounded and responsible AI** — responses are based on the organization's travel knowledge base and protected by input/output safety controls.
- **Evaluation-driven development** — RAGAS provides measurable validation of retrieval and response quality, with iterative tuning of chunking and retrieval parameters.
- **LLMOps integrated throughout the lifecycle** — experiment tracking, evaluation artifacts, metrics, and reproducible ingestion support continuous improvement.
- **Containerized application** provides a consistent deployment model from local development to Azure.
- **Repeatable and maintainable architecture** separates data storage, retrieval, generation, governance, evaluation, and application layers, making the solution easier to extend and operate.

Expected Business Impact

Improve customer experience with fast, consistent, 24/7 answers to travel and airline policy questions.

Reduce operational workload by automating repetitive travel-support queries and allowing support teams to focus on complex cases.

Increase response accuracy and trust through authoritative source documents, grounded generation, safety controls, and measurable evaluation.

Reduce business risk associated with hallucinated, unsafe, or outdated responses.

Accelerate knowledge updates by allowing new and updated policy documents to flow through the ingestion pipeline into the searchable knowledge base.

Enable continuous optimization using RAGAS metrics and MLflow experiment tracking rather than relying solely on subjective testing.

Provide a scalable foundation for expanding from static policy Q&A to real-time travel disruptions, cancellations, refunds, schedule changes, and other customer-service scenarios.



Power Ahead!

